Unit 10. Autoregressive Models and Forecasting

1 Autoregressive Models of Order One

FREES §8.2 (SRM reading list; not on the PA list). SRM learning outcomes 3(a), 3(b).

Definition 10.1 (White noise and the innovations)**.** Throughout (except in the section on ARCH/GARCH models), $\{\varepsilon_t\}_{t\in\mathbb{Z}}$ is independent and identically distributed with mean 0 and variance $\sigma^2 \in (0,\infty)$, and $\mathcal{L}_t$ is the prerequisites' **past space**: the closure in L^2 of the square-integrable functions of finitely many ε_u with $u \leqslant t$. For $X \in L^2$, $\mathsf{E}[X \mid \mathcal{L}_t]$ denotes the orthogonal projection of X on the closed subspace $\mathcal{L}_t$: the nearest-point argument in the proof of the prerequisites' theorem **conditional expectation in** L^2 applies to $\mathcal{L}_t$, and the error $X - \mathsf{E}[X \mid \mathcal{L}_t]$ is orthogonal to $\mathcal{L}_t$.

Definition 10.2 (Autoregressive model of order one; FREES (8.1))**.** The AR(1) model is

$$y_t = \beta_0 + \beta_1 y_{t-1} + \varepsilon_t, \tag{1}$$

with $|\beta_1| < 1$ and ε_t independent of $\mathcal{L}_{t-1}$. FREES requires instead $\operatorname{cov}(\varepsilon_{t+k}, y_t) = 0$ for $k > 0$, which follows because the solution lies in $\mathcal{L}_t$, as this section proves.

Theorem 10.3 (Existence, uniqueness, moments; FREES §8.2, exercise 8.4)**.** Let $|\beta_1| < 1$ and $\mu = \beta_0/(1-\beta_1)$. Then

$$y_t = \mu + \sum_{k=0}^{\infty} \beta_1^k \, \varepsilon_{t-k} \tag{2}$$

converges in L^2, satisfies (1), and is the only weakly stationary solution of (1). It is weakly stationary with

$$\mathsf{E}[y_t] = \mu, \qquad \gamma(0) = \operatorname{var}(y_t) = \frac{\sigma^2}{1-\beta_1^2}, \qquad \gamma(k) = \beta_1^k \gamma(0), \qquad \rho_k = \beta_1^k. \tag{3}$$

Proof. The coefficients β_1^k are square summable, so (2) converges in L^2 and lies in $\mathcal{L}_t$ by the prerequisites' construction of the past space. Splitting off the first term,

$$y_t = \mu + \varepsilon_t + \beta_1 \sum_{k=0}^{\infty} \beta_1^k \varepsilon_{t-1-k} = \mu + \varepsilon_t + \beta_1 (y_{t-1} - \mu),$$

and $\mu - \beta_1 \mu = \beta_0$, so (1) holds.

For the moments, $\mathsf{E}[y_t] = \mu$ because each ε has mean zero and L^2 convergence preserves the mean. The innovations are uncorrelated with common variance σ^2, so for $k \geqslant 0$

$$\gamma(k) = \operatorname{cov}\Big(\sum_{j\geqslant 0} \beta_1^j \varepsilon_{t-j},\ \sum_{l\geqslant 0} \beta_1^l \varepsilon_{t+k-l}\Big) = \sigma^2 \sum_{j\geqslant 0} \beta_1^j \beta_1^{j+k} = \sigma^2 \beta_1^k \sum_{j\geqslant 0} \beta_1^{2j} = \frac{\sigma^2 \beta_1^k}{1-\beta_1^2},$$

the pairing $j = l - k$ being the only one that survives. Both moments are free of t.
For uniqueness, let $\{\tilde{y}_t\}$ be another weakly stationary solution and put $d_t = y_t - \tilde{y}_t$. Subtracting the two copies of (1) gives $d_t = \beta_1 d_{t-1} = \beta_1^m d_{t-m}$ for every m. Weak stationarity bounds $\mathsf{E}[d_{t-m}^2]$ by a constant C free of m, so $\mathsf{E}[d_t^2] \leqslant \beta_1^{2m} C \to 0$, forcing $d_t = 0$ almost surely. □

Remark (FREES §8.2)**.** A correlogram decaying geometrically identifies an AR(1) (FREES (8.2)); FREES calls a series with $\rho_1 > 0$ **meandering**. By (3), $\gamma(0)(1-\beta_1^2) = \sigma^2$: for $\beta_1 \neq 0$ the series is more variable than its innovations, increasingly so as $\beta_1 \to 1$.

Proposition 10.4 (The two boundary cases; FREES §8.2)**.** Setting $\beta_1 = 0$ in (1) gives white noise about β_0. Setting $\beta_1 = 1$ gives $\Delta y_t = \beta_0 + \varepsilon_t$, a random walk with drift β_0, for which no

weakly stationary solution exists.

Proof. The first is immediate. For the second, $\beta_1 = 1$ turns (1) into $y_t - y_{t-1} = \beta_0 + \varepsilon_t$, which is Unit 9's random walk. If a weakly stationary solution existed, iterating would give $y_t - y_{t-m} = m\beta_0 + \sum_{j<m} \varepsilon_{t-j}$ for every $m \geqslant 1$, whose variance is $m\sigma^2$ because the ε_t are iid; but for a weakly stationary series $\operatorname{var}(y_t - y_{t-m}) = 2\gamma(0) - 2\gamma(m) \leqslant 4\gamma(0)$ for every m, which is impossible since $\sigma^2 > 0$. □

Proposition 10.5 (Fourth moment)**.** If in addition $\mu_4 = \mathsf{E}[\varepsilon_t^4] < \infty$, then $x_t = y_t - \mu$ satisfies

$$\mathsf{E}[x_t^4] = \frac{\mu_4 + 6\beta_1^2\sigma^2\gamma(0)}{1 - \beta_1^4} < \infty, \tag{4}$$

and $\{x_t^2\}$ is weakly stationary with autocovariance $\operatorname{cov}(x_t^2, x_{t+k}^2) = \beta_1^{2k} \operatorname{var}(x_t^2) \to 0$. Consequently $T^{-1} \sum_{t \leqslant T} x_t^2 \xrightarrow{p} \gamma(0)$.

Proof. Finiteness. For the partial sum $S_m = \sum_{j<m} \beta_1^j \varepsilon_{t-j}$, expanding the fourth power and using independence with $\mathsf{E}[\varepsilon] = 0$ leaves only the terms in which every index occurs at least twice: the diagonal, and the three ways of splitting four factors into two distinct pairs. Hence

$$\mathsf{E}[S_m^4] = \mu_4 \sum_{j<m} \beta_1^{4j} + 3\sigma^4 \Big\{ \Big(\sum_{j<m} \beta_1^{2j} \Big)^2 - \sum_{j<m} \beta_1^{4j} \Big\} \leqslant \frac{\mu_4}{1-\beta_1^4} + \frac{3\sigma^4}{(1-\beta_1^2)^2},$$

a bound free of m. Now $S_m \to x_t$ in L^2, hence in probability and so almost surely along a subsequence, and Fatou's lemma gives $\mathsf{E}[x_t^4] < \infty$. The law of x_t is free of t, being the weak limit of the laws of the S_m, which are.

Value. Write $m_4 = \mathsf{E}[x_t^4]$ and use $x_t = \beta_1 x_{t-1} + \varepsilon_t$ with ε_t independent of x_{t-1}. The two cross terms with coefficient 4 in the expansion carry a factor $\mathsf{E}[\varepsilon_t] = 0$ or $\mathsf{E}[\varepsilon_t^3]\,\mathsf{E}[x_{t-1}] = 0$, and the term $6\beta_1^2 x_{t-1}^2 \varepsilon_t^2$ has expectation $6\beta_1^2\sigma^2\,\mathsf{E}[x_{t-1}^2]$, so $m_4 = \beta_1^4 m_4 + 6\beta_1^2\sigma^2\,\mathsf{E}[x_{t-1}^2] + \mu_4$, which is (4).

Autocovariance. Iterate (1) k times to get $x_{t+k} = \beta_1^k x_t + v$ with $v = \sum_{j<k} \beta_1^j \varepsilon_{t+k-j}$ independent of x_t and of mean zero. Then $x_{t+k}^2 = \beta_1^{2k} x_t^2 + 2\beta_1^k x_t v + v^2$, and $\operatorname{cov}(x_t^2, x_t v) = \mathsf{E}[x_t^3]\,\mathsf{E}[v] - \mathsf{E}[x_t^2]\,\mathsf{E}[x_t]\,\mathsf{E}[v] = 0$ while $\operatorname{cov}(x_t^2, v^2) = 0$, both by independence. So $\operatorname{cov}(x_t^2, x_{t+k}^2) = \beta_1^{2k} \operatorname{var}(x_t^2)$, which tends to 0, and the prerequisites' lemma **variance of a stationary average** applies. □

Remark (FREES §8.2; SOA sample questions 22 and 64)**.** Stationarity restricts β_1 to $(-1, 1)$ and leaves β_0 free (Theorem 10.3); $\beta_1 = 0$ is white noise and $\beta_1 = 1$ a random walk (Proposition 10.4).

Theorem 10.6 (Joint limit of the sample autocorrelations; FREES §8.2)**.** Let $y_1, \dots, y_T$ be white noise with mean μ, variance σ^2 and $\mathsf{E}\, y_t^4 < \infty$, and fix $K \geqslant 1$. Then

$$\sqrt{T}\,(r_1, \dots, r_K)' \xrightarrow{d} N(\mathbf{0},\ \mathbf{I}_K). \tag{5}$$

Proof. Put $\xi_t = y_t - \mu$ and $\mathbf{a}_j = \mathbf{e}_{j+1} \in \mathbb{R}^K$ for $0 \leqslant j \leqslant K-1$, $\mathbf{a}_j = \mathbf{0}$ beyond, so that $\mathbf{u}_{t-1} = \sum_j \mathbf{a}_j \xi_{t-1-j} = (\xi_{t-1}, \dots, \xi_{t-K})'$ and $\mathbf{\Gamma} = \mathsf{E}[\mathbf{u}\mathbf{u}'] = \sigma^2 \mathbf{I}_K$. The prerequisites' central limit theorem for autoregressive scores gives

$$\frac{1}{\sqrt{T}} \sum_{t=K+1}^{T} \mathbf{u}_{t-1}\xi_t \xrightarrow{d} N(\mathbf{0},\ \sigma^4 \mathbf{I}_K),$$

the theorem's sum running from $t = 1$ and this one from $t = K+1$, a difference of K terms $T^{-1/2}\mathbf{u}_{t-1}\xi_t$, each of which tends to 0 in probability by Chebyshev's inequality, its expected squared

norm being $K\sigma^4/T$.

The kth coordinate of the sum is $\sum_t \xi_t \xi_{t-k}$, the numerator of r_k with the mean known, up to the $K-k$ terms with $k < t \leqslant K$, which after division by $\sqrt{T}$ tend to 0 in probability by the same Chebyshev argument. Writing $\bar{\xi} = \bar{y} - \mu$, the numerator actually used differs from it by

$$-\bar{\xi} \sum_t \xi_{t-k} - \bar{\xi} \sum_t \xi_t + (T-k)\bar{\xi}^2,$$

and after division by $\sqrt{T}$ each of the three terms is the product of $\sqrt{T}\,\bar{\xi}$, which converges in distribution to $N(0, \sigma^2)$ by the prerequisites' central limit theorem for independent summands, with a factor that tends to 0 in probability ($T^{-1}\sum_t \xi_{t-k}$, $T^{-1}\sum_t \xi_t$ or $(T-k)T^{-1}\bar{\xi}$, by the lemma cited next), so each tends to 0 in probability by Slutsky's theorem. The denominator satisfies $T^{-1}\sum_t (y_t - \bar{y})^2 = T^{-1}\sum_t \xi_t^2 - \bar{\xi}^2 \xrightarrow{p} \sigma^2$, by the lemma **variance of a stationary average** of the prerequisites applied to the iid sequence $\{\xi_t^2\}$, which is square integrable because $\mathsf{E}\, y_t^4 < \infty$, and to $\{\xi_t\}$, which gives $\bar{\xi} \xrightarrow{p} 0$. Dividing and applying Slutsky's theorem gives (5). □

Corollary 10.7 (The correlogram band)**.** Under white noise with $\mathsf{E}\, y_t^4 < \infty$, $\mathsf{P}\{|r_k| > 2/\sqrt{T}\} \to \mathsf{P}\{|Z| > 2\} = 0.0455$ for each fixed k, Z being standard normal.

Proof. $\sqrt{T}\, r_k \xrightarrow{d} N(0,1)$ by Theorem 10.6, and ± 2 are continuity points of the normal distribution function. □

Remark (FREES §8.2)**.** FREES's standard error $\mathrm{se}(r_k) = 1/\sqrt{T}$ under no autocorrelation and its rule "$|r_k| > 2/\sqrt{T}$" are Corollary 10.7: a test of level 4.55% in the limit.

Example 10.1 (Inflation bond index; FREES §8.2, Table 8.4)**.** FREES's $T = 51$ monthly values (Unit 9):

```
yb <- read.csv("data/frees/csv/InflationBond.csv")$INFBOND; Tb <- length(yb)
rb <- acf(yb, lag.max = 5, plot = FALSE)$acf[2:6]
round(rbind(r_k = rb, ar1_r1_power_k = rb[1]^(1:5)), 3)
```

```
##                  [,1]  [,2]  [,3]  [,4]  [,5]
## r_k             0.814 0.632 0.561 0.447 0.367
## ar1_r1_power_k 0.814 0.662 0.539 0.439 0.357
```

```r
round(c(se = 1 / sqrt(Tb), r1_over_se = rb[1] * sqrt(Tb)), 3)
```

```
##          se r1_over_se
##       0.140      5.812
```

$r_1 = 0.814$ is 5.81 standard errors from 0, so white noise is rejected. Under an AR(1) the estimates r_1^k of ρ_k track the r_k within 0.03 at every lag; FREES's Table 8.4 lists $r_5 = 0.267$, which is 0.367 (Unit 9).

2 Estimation and Diagnostic Checking

FREES §8.3. SRM learning outcomes 3(b), 3(c).

Definition 10.8 (Conditional least squares; FREES §8.3)**.** Given $y_1, \ldots, y_T$, the **conditional least squares** estimates b_0, b_1 minimise $\sum_{t=2}^{T}(y_t - b_0 - b_1 y_{t-1})^2$: least squares of y_t on y_{t-1} over the $T-1$ usable dates. The residuals are $e_t = y_t - (b_0 + b_1 y_{t-1})$ and

$$s^2 = \frac{1}{T-3}\sum_{t=2}^{T}(e_t - \bar{e})^2. \tag{6}$$

Proposition 10.9 (Exact and approximate estimates; FREES §8.3, exercise 8.3)**.** With $\bar{y}_- = (T-1)^{-1}\sum_{t=1}^{T-1} y_t$ and $\bar{y}_+ = (T-1)^{-1}\sum_{t=2}^{T} y_t$,

$$b_1 = \frac{\sum_{t=2}^{T}(y_{t-1}-\bar{y}_-)(y_t-\bar{y}_+)}{\sum_{t=2}^{T}(y_{t-1}-\bar{y}_-)^2}, \qquad b_0 = \bar{y}_+ - b_1\bar{y}_-, \tag{7}$$

and $\sum_t e_t = 0$, so $\bar{e} = 0$ and (6) is the residual sum of squares over $T-3$. If $y_1 = y_T = \bar{y}$, then $b_1 = r_1$ and $b_0 = \bar{y}(1-r_1)$; in general both differences are end effects (not quantified here).

Proof. The formulas are Unit 2's simple regression applied to the $T-1$ pairs (y_{t-1}, y_t), whose residuals sum to zero because of the intercept. If $y_1 = y_T = \bar{y}$, then $\bar{y}_- = (T\bar{y} - y_T)/(T-1) = \bar{y}$ and likewise $\bar{y}_+ = (T\bar{y} - y_1)/(T-1) = \bar{y}$, so the numerator of b_1 is $\sum_{t\geqslant 2}(y_{t-1}-\bar{y})(y_t-\bar{y})$, the numerator of r_1, and its denominator $\sum_{t\leqslant T-1}(y_t-\bar{y})^2 = \sum_{t\leqslant T}(y_t-\bar{y})^2$, that of r_1; then $b_0 = \bar{y} - r_1\bar{y}$. □

Remark (FREES (8.3))**.** FREES's $\bar{e}$ in (6), "because without the first residual the average of the residuals is no longer automatically zero", is 0 for the least squares fit by Proposition 10.9; it is not

zero for the approximation $b_1 = r_1$, $b_0 = \bar{y}(1 - r_1)$, which is what SOA sample question 58 uses.

Theorem 10.10 (Asymptotics of conditional least squares; FREES §8.3)**.** Let $\{y_t\}$ be the stationary AR(1) of Theorem 10.3 with $\mathsf{E}[\varepsilon_t^4] < \infty$. Then, as $T \to \infty$,

$$b_1 \xrightarrow{p} \beta_1, \qquad \sqrt{T}\,(b_1 - \beta_1) \xrightarrow{d} N\big(0,\ 1 - \beta_1^2\big). \tag{8}$$

Proof. Write $x_t = y_t - \mu$, so that (7) becomes, after subtracting μ from every observation,

$$b_1 - \beta_1 = \frac{\sum_{t=2}^{T}(x_{t-1} - \bar{x}_{-})\varepsilon_t}{\sum_{t=2}^{T}(x_{t-1} - \bar{x}_{-})^2},$$

because $x_t = \beta_1 x_{t-1} + \varepsilon_t$ makes the numerator of (7) equal to $\beta_1 \sum (x_{t-1} - \bar{x}_{-})^2 + \sum (x_{t-1} - \bar{x}_{-})\varepsilon_t$, the centring of the second factor contributing nothing since $\sum_t (x_{t-1} - \bar{x}_{-}) = 0$.

Denominator. $\{x_t\}$ is weakly stationary with $\gamma(k) = \beta_1^k \gamma(0) \to 0$, so $\bar{x}_{-} \xrightarrow{p} 0$ by the prerequisites' lemma **variance of a stationary average**, and $\{x_t^2\}$ is weakly stationary with vanishing

autocovariance by Proposition 10.5, so $T^{-1}\sum_t x_{t-1}^2 \xrightarrow{p} \gamma(0)$. Hence

$$\frac{1}{T}\sum_{t=2}^{T}(x_{t-1}-\bar{x}_{-})^2 = \frac{1}{T}\sum_{t=2}^{T}x_{t-1}^2 - \frac{T-1}{T}\bar{x}_{-}^2 \xrightarrow{p} \gamma(0) > 0.$$

Numerator. With $a_j = \beta_1^j$ the sequence $x_{t-1} = \sum_{j\geqslant 0} a_j \varepsilon_{t-1-j}$ is exactly the linear combination of the prerequisites' central limit theorem for autoregressive scores, whose limit variance is $\sigma^2 \cdot \sigma^2 \sum_j \beta_1^{2j} = \sigma^2\gamma(0)$. So $T^{-1/2}\sum_t x_{t-1}\varepsilon_t \xrightarrow{d} N(0, \sigma^2\gamma(0))$, the theorem's sum running from $t=1$ and this one from $t=2$, a difference $x_0\varepsilon_1/\sqrt{T}$ that tends to 0 in probability by Chebyshev's inequality. The centring costs nothing:

$$\frac{1}{\sqrt{T}}\sum_{t=2}^{T}(x_{t-1}-\bar{x}_{-})\varepsilon_t = \frac{1}{\sqrt{T}}\sum_{t=2}^{T}x_{t-1}\varepsilon_t - \bar{x}_{-}\cdot\frac{1}{\sqrt{T}}\sum_{t=2}^{T}\varepsilon_t,$$

and the second term is a product of $\bar{x}_{-} \xrightarrow{p} 0$ with a quantity converging in distribution to $N(0,\sigma^2)$ by the prerequisites' central limit theorem for independent summands, hence tends to 0 in probability by Slutsky's theorem.

Ratio. Slutsky's theorem again gives $\sqrt{T}(b_1 - \beta_1) \xrightarrow{d} N(0, \sigma^2\gamma(0)/\gamma(0)^2)$, and $\sigma^2/\gamma(0) = 1 - \beta_1^2$ by (3). Consistency follows from the same two limits with the $\sqrt{T}$ removed.

Intercept and residual variance. Since $\beta_0 = \mu(1 - \beta_1)$, (7) gives $b_0 - \beta_0 = \bar{x}_+ - b_1\bar{x}_- - \mu(b_1 - \beta_1)$, with $\bar{x}_\pm = \bar{y}_\pm - \mu$. By the lemma **variance of a stationary average**, $T \operatorname{var}(\bar{x}_\pm) \leqslant c\sum_k |\gamma(k)| < \infty$ for a constant c not depending on T, so by Chebyshev's inequality $\mathsf{P}\{\sqrt{T}|\bar{x}_\pm| > M\} \leqslant c\sum_k |\gamma(k)|/M^2$ for every T; together with $b_1 \xrightarrow{p} \beta_1$ and (8), this gives $b_0 \xrightarrow{p} \beta_0$ and makes $\mathsf{P}\{\sqrt{T}|b_0 - \beta_0| > M\}$ small, uniformly in large T, for M large: the error in b_0 is of order $T^{-1/2}$, as is that in b_1. Next, $e_t = \varepsilon_t - (b_0 - \beta_0) - (b_1 - \beta_1)y_{t-1}$, so $T^{-1}\sum_t e_t^2 - T^{-1}\sum_t \varepsilon_t^2$ is a sum of products, each of a factor $b_0 - \beta_0$ or $b_1 - \beta_1$ (which tend to 0 in probability) with one of, or a product of two of, $b_0 - \beta_0$, $b_1 - \beta_1$, $T^{-1}\sum_t \varepsilon_t$, $T^{-1}\sum_t y_{t-1}$, $T^{-1}\sum_t y_{t-1}\varepsilon_t$ and $T^{-1}\sum_t y_{t-1}^2$ (each of which converges in probability to a constant, by the limits above); hence it tends to 0 in probability. Finally $T^{-1}\sum_t \varepsilon_t^2 \xrightarrow{p} \sigma^2$ by the same lemma applied to the iid, square-integrable $\{\varepsilon_t^2\}$, and $T/(T-3) \to 1$, so $s^2 \xrightarrow{p} \sigma^2$. □

Corollary 10.11 (Standard error)**.** An approximate standard error for b_1 is $\sqrt{(1-b_1^2)/T}$, and its ratio to the ordinary least squares standard error $s/\{\sum_t (y_{t-1}-\bar{y}_-)^2\}^{1/2}$ reported by a regression routine tends to 1 in probability.

Proof. The first expression is (8) with β_1 estimated. For the second, the regression standard error squared is $s^2/\sum_t (y_{t-1}-\bar{y}_-)^2$, whose numerator converges in probability to σ^2 (proof of Theorem 10.10) and whose denominator is T times a quantity converging to $\gamma(0)$; the ratio times T therefore tends to $\sigma^2/\gamma(0) = 1-\beta_1^2$, which is also the limit of T times the square of the first expression. The ratio of the two standard errors therefore tends to 1. □

Example 10.2 (Inflation bond index, fitted; FREES §8.3, Table 8.5)**.**

```
mb <- lm(yb[-1] ~ yb[-Tb]); eb <- residuals(mb)
round(coef(summary(mb))[, 1:3], 4)
```

```
##             Estimate Std. Error t value
## (Intercept)   0.4085     0.1692   2.415
## yb[-Tb]       0.8138     0.0749  10.872
```

```
round(c(s = sqrt(sum((eb - mean(eb))^2) / (Tb - 3)), se_asymptotic = sqrt((1 -
 ↪  coef(mb)[[2]]^2) / Tb),
       b0_approx = mean(yb) * (1 - rb[1])), 4)
```

```
##             s se_asymptotic     b0_approx
##        0.1370        0.0814        0.4179
```

```
am <- arima(yb, order = c(1, 0, 0), method = "ML")
round(c(ml_ar1 = coef(am)[["ar1"]], ml_b0 = coef(am)[["intercept"]] * (1 -
 ↪  coef(am)[["ar1"]])), 4)
```

```
## ml_ar1  ml_b0
## 0.8538 0.3347
```

```
round(rbind(cls_residuals = acf(eb, lag.max = 5, plot = FALSE)$acf[2:6],
            ml_residuals = acf(residuals(am), lag.max = 5, plot = FALSE)$acf[2:6]),
             ↪  3)
```

```
##                [,1]   [,2]  [,3]   [,4]   [,5]
## cls_residuals 0.157 -0.289 0.059  0.073 -0.124
## ml_residuals  0.082 -0.294 0.103 -0.006 -0.164
```

Conditional least squares gives $0.4085 + 0.8138\, y_{t-1}$, standard errors 0.1692 and 0.0749, $s = 0.137$.

FREES reports $0.2923 + 0.8727\, y_{t-1}$ with standard errors 0.0196 and 0.0736 and $s = 0.14$; neither conditional least squares nor exact maximum likelihood $(0.3347 + 0.8538\, y_{t-1})$ reproduces it, and an intercept standard error of 0.0196 is not possible here, the series having mean 2.245 and the slope standard error 0.0736. FREES's residual autocorrelations (0.09, −0.33, 0.07, 0.02, −0.17) are close to those of the maximum likelihood fit. In both, $r_2(e)$ is about −2 standard errors $1/\sqrt{51} = 0.14$. Figure 10.1 shows the fitted line and the control chart of the residuals.

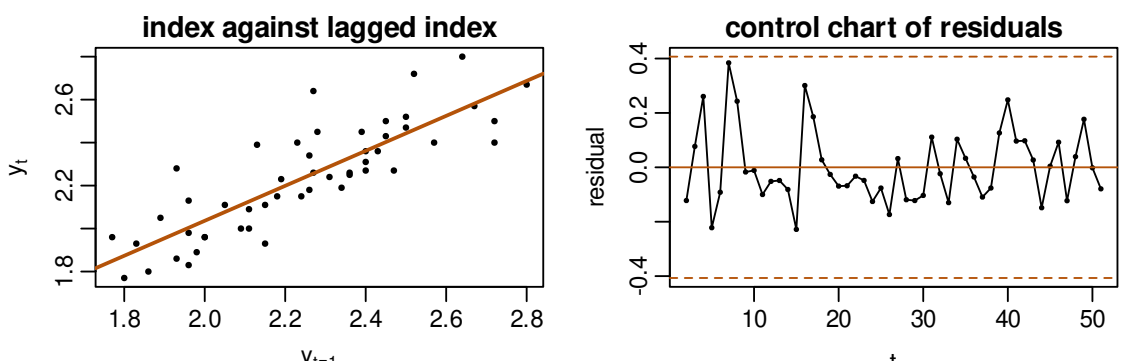


Figure 10.1: Inflation bond index (FREES Figures 8.2 and 8.4): the index against its lagged value with the conditional least squares line, and the control chart of the residuals.

Remark (Residual autocorrelations; FREES §8.3, §8.6)**.** Under a fitted AR(1) the standard error $1/\sqrt{T}$ of $r_k(e)$ is only approximate; for $k = 1$ it is nearer $|\beta_1|/\sqrt{T}$ (FREES §8.6, citing MacLeod), and that refinement is not proved here.

Definition 10.12 (Portmanteau statistics (beyond the syllabus); FREES §8.6)**.** For residual autocorrelations $r_k(e)$ and a user-chosen K,

$$Q_{\mathrm{BP}} = T\sum_{k=1}^{K} r_k(e)^2, \qquad Q_{\mathrm{LB}} = T(T+2)\sum_{k=1}^{K} \frac{r_k(e)^2}{T-k}. \tag{9}$$

Theorem 10.13 (Limit for an unfitted series (beyond the syllabus))**.** If $y_1, \ldots, y_T$ is white noise with $\mathsf{E}\, y_t^4 < \infty$ and the r_k are computed from the series itself, then $Q_{\mathrm{BP}} \xrightarrow{d} \chi^2_K$ and $Q_{\mathrm{LB}} \xrightarrow{d} \chi^2_K$.

Proof. $Q_{\mathrm{BP}} = |\sqrt{T}\,\mathbf{r}|^2$ with $\mathbf{r} = (r_1, \ldots, r_K)'$, and Theorem 10.6 with the continuous mapping

theorem gives $|\sqrt{T}\mathbf{r}|^2 \xrightarrow{d} |Z|^2$ for $Z \sim N(\mathbf{0}, \mathbf{I}_K)$, which is χ^2_K by the prerequisites. For Q_{LB}, write

$$Q_{\mathrm{LB}} = \sum_{k=1}^{K} \frac{T(T+2)}{T-k} r_k^2 = \sum_{k=1}^{K} \Big(1 + \frac{k+2}{T-k}\Big) T\, r_k^2,$$

so $Q_{\mathrm{LB}} - Q_{\mathrm{BP}} = \sum_k \{(k+2)/(T-k)\} T r_k^2 \xrightarrow{p} 0$, because $T r_k^2 \xrightarrow{d} \chi^2_1$ by Theorem 10.6 and the continuous mapping theorem while $(k+2)/(T-k) \to 0$, so each product tends to 0 in probability by Slutsky's theorem; Slutsky's theorem finishes. □

Remark. When the r_k are computed from the residuals of a fitted ARMA(p, q), the limit is χ^2_{K-p-q} rather than χ^2_K: fitting removes exactly $p + q$ directions from the residual correlogram. That statement is **not** proved here. It needs the joint limit of the residual autocorrelations, which involves the estimation error and is beyond the prerequisites as they stand. For the AR(1) it is measured by simulation in this unit instead: the mean of Q_{LB} sits near $K - 1 = K - p - q$, and the size is the nominal 5% only when the degrees of freedom are reduced.

Remark (FREES §8.6)**.** FREES refers Q_{BP} and Q_{LB} computed from the residuals of a fitted ARMA(p,q) to χ^2 with K minus the number of linear parameters, $1+p+q$; the usual reference subtracts $p+q$. Neither statement for fitted residuals is proved here: it needs the joint law of the residual autocorrelations, which carries the estimation error.

3 Smoothing and Prediction

FREES §8.4. SRM learning outcomes 3(b), 3(c).

Definition 10.14 (Smoothed series and chain rule; FREES §8.4, (8.4))**.** The **smoothed series** is $\hat{y}_t = b_0 + b_1 y_{t-1}$. The k-step forecast is $\hat{y}_{T+k} = b_0 + b_1 \hat{y}_{T+k-1}$, starting from $\hat{y}_T = y_T$: the **chain rule of forecasting**.

Theorem 10.15 (Optimal forecast and its error; FREES §8.4, exercise 8.5)**.** For the stationary AR(1) with known parameters, the minimum mean squared error predictor of y_{T+l} among all square-integrable functions of the past is

$$\mathsf{E}[y_{T+l} \mid \mathcal{L}_T] = \mu + \beta_1^l (y_T - \mu), \tag{10}$$

and its error variance is

$$\operatorname{var}(y_{T+l} - \mathsf{E}[y_{T+l} \mid \mathcal{L}_T]) = \sigma^2 \sum_{k=0}^{l-1} \beta_1^{2k} = \sigma^2 \frac{1 - \beta_1^{2l}}{1 - \beta_1^2}. \tag{11}$$

The point forecast tends to μ and the error variance to $\gamma(0)$ as $l \to \infty$.

Proof. Iterating (1) forward l steps,

$$y_{T+l} - \mu = \beta_1^l (y_T - \mu) + \sum_{k=0}^{l-1} \beta_1^k \varepsilon_{T+l-k}. \tag{12}$$

Each innovation in the sum has index exceeding T, hence is independent of $\mathcal{L}_T$ and has mean zero, while $y_T - \mu \in \mathcal{L}_T$; taking conditional expectations gives (10), which minimises mean squared error by the projection property of conditional expectation in L^2 (prerequisites). The forecast error is the sum in (12), whose terms are uncorrelated with variances $\sigma^2 \beta_1^{2k}$, giving (11). The limits follow from $|\beta_1| < 1$ and (3). □

Corollary 10.16 (Forecast interval)**.** If moreover the ε_t are normal, an approximate $100(1-\alpha)\%$ interval for y_{T+l} is

$$\hat{y}_{T+l} \;\pm\; t_{T-3,\,1-\alpha/2}\; s\,\Big(\sum_{k=0}^{l-1} b_1^{2k}\Big)^{1/2}, \tag{13}$$

which for $l=1$ is $\hat{y}_{T+1} \pm t\,s$ and for $l=2$ is $\hat{y}_{T+2} \pm t\,s(1+b_1^2)^{1/2}$.

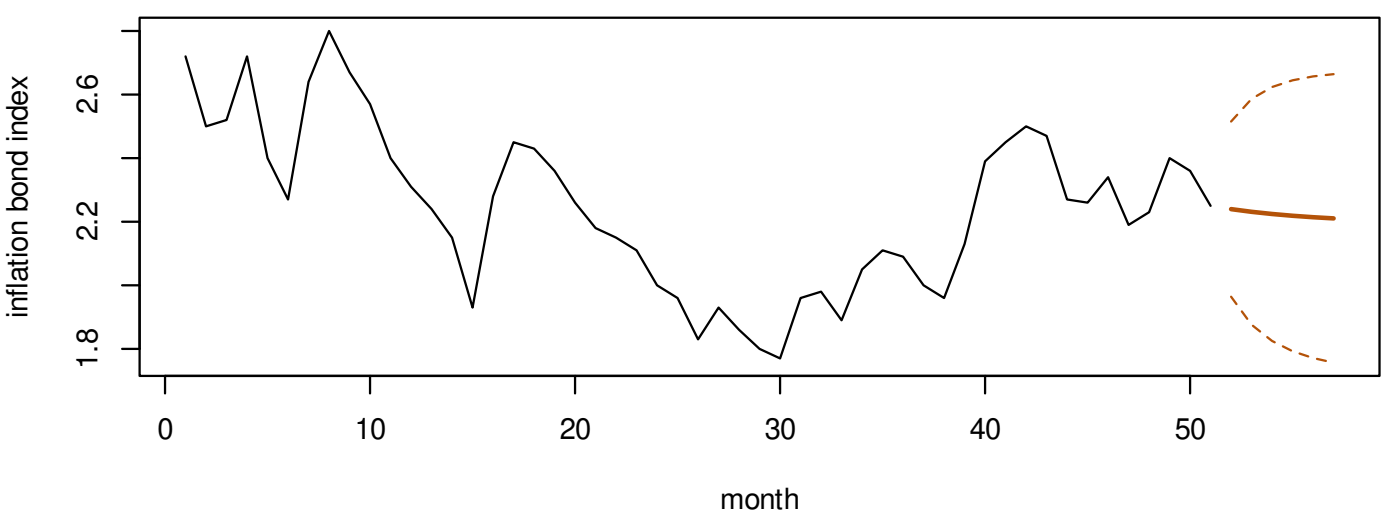
Proof. With known parameters the forecast error is the sum in (12), a fixed linear combination of l innovations; for normal ε_t it is $N\big(0, \sigma^2 \sum_{k<l} \beta_1^{2k}\big)$, so the interval with β_0, β_1, σ in place of b_0, b_1, s and the normal quantile $z_{1-\alpha/2}$ in place of t covers y_{T+l} with probability exactly $1-\alpha$. Substitute the estimates into (11) and treat the estimation error in b_0, b_1 as negligible: by Theorem 10.10 and its proof, b_0, b_1 err by order $T^{-1/2}$ and $s \xrightarrow{p} \sigma$, so the estimated centre differs from the known-parameter one by a term tending to 0 in probability, the half-width converges in probability to the known-parameter one, and $t_{T-3,1-\alpha/2} \to z_{1-\alpha/2}$; by Slutsky's theorem the coverage tends to $1-\alpha$. Without normality the law of the error does not depend on T and is not normal in general, so the coverage need not approach $1-\alpha$. □

Remark (FREES §8.4). FREES's interval is $\hat{y}_{T+k} \pm t\, s\{1 + b_1^2 + \cdots + b_1^{2(k-1)}\}^{1/2}$ with t from the t_{T-3} law, "the percentile $1 - (\text{prediction level})/2$"; the percentile is $1 - (1 - \text{level})/2$, that is 0.975 for a 95% interval.

Example 10.3 (Forecasting the inflation bond index; FREES §8.4, Figure 8.6).

```
b0b <- coef(mb)[[1]]; b1b <- coef(mb)[[2]]; sb <- summary(mb)$sigma
fcb <- Reduce(function(prev, k) b0b + b1b * prev, 1:6, accumulate = TRUE, init =
 ↪  yb[Tb])[-1]
round(cbind(k = 1:6, forecast = fcb, half_width = qt(0.975, Tb - 3) * sb *
              sqrt(cumsum(b1b^(2 * (0:5)))), limit_mean = b0b / (1 - b1b),
            limit_half = qt(0.975, Tb - 3) * sb / sqrt(1 - b1b^2)), 4)
```

```
##      k forecast half_width limit_mean limit_half
## [1,] 1    2.240     0.2756      2.194     0.4742
## [2,] 2    2.231     0.3553      2.194     0.4742
## [3,] 3    2.224     0.3994      2.194     0.4742
## [4,] 4    2.219     0.4261      2.194     0.4742
## [5,] 5    2.214     0.4429      2.194     0.4742
## [6,] 6    2.211     0.4537      2.194     0.4742
```



From $y_{51} = 2.25$ the forecasts move towards $b_0/(1-b_1) = 2.194$, and the half-widths grow towards $t\,s/\sqrt{1-b_1^2}$, the limits of Theorem 10.15. The plot, FREES's Figure 8.6, shows the series with the six forecasts and their 95% intervals.

4 Box–Jenkins Models (beyond the syllabus)

Beyond the syllabus: FREES §8.5–8.6, outside the SRM list (chapter 8, sections 1–4), and ISLR §10.5.2 with the lab §10.9.6, on neither list.

Definition 10.17 (Backshift; FREES §8.5.1)**.** $\mathrm{B}y_t = y_{t-1}$ and $\mathrm{B}^k y_t = y_{t-k}$. For a polynomial $\Phi(x) = 1 - \beta_1 x - \cdots - \beta_p x^p$, $\Phi(\mathrm{B})y_t = y_t - \beta_1 y_{t-1} - \cdots - \beta_p y_{t-p}$.

Definition 10.18 (AR(p), MA(q), ARMA, ARIMA; FREES (8.5)–(8.10))**.** With $\Phi(x) = 1 - \beta_1 x - \cdots - \beta_p x^p$ and $\Theta(x) = 1 - \theta_1 x - \cdots - \theta_q x^q$,

$$\mathrm{AR}(p): \quad \Phi(\mathrm{B})y_t = \beta_0 + \varepsilon_t, \tag{14}$$

$$\mathrm{MA}(q): \quad y_t = \beta_0 + \Theta(\mathrm{B})\varepsilon_t, \tag{15}$$

$$\mathrm{ARMA}(p,q): \quad \Phi(\mathrm{B})y_t = \beta_0 + \Theta(\mathrm{B})\varepsilon_t, \tag{16}$$

and ARIMA(p,d,q) is ARMA(p,q) applied to $w_t = (1-\mathrm{B})^d y_t$.

Proposition 10.19 (MA(q) is always stationary and its correlogram cuts off)**.** The MA(q) process (15) is weakly stationary for every $\theta_1, \dots, \theta_q$, with $\theta_0 = -1$ and

$$\gamma(k) = \sigma^2 \sum_{j=0}^{q-k} \theta_j \theta_{j+k} \quad (0 \leqslant k \leqslant q), \qquad \gamma(k) = 0 \quad (k > q). \tag{17}$$

Proof. $y_t - \beta_0$ is a finite linear combination of $\varepsilon_t, \dots, \varepsilon_{t-q}$ with coefficients $-\theta_j$ and $\theta_0 = -1$, so the mean is β_0 and the covariance is $\sigma^2 \sum_j \theta_j \theta_{j+k}$ over those j with both indices in $\{0, \dots, q\}$, all free of t. For $k > q$ the two windows $\{t-q, \dots, t\}$ and $\{t+k-q, \dots, t+k\}$ are disjoint and the innovations independent. □

Theorem 10.20 (Stationarity of AR(p))**.** Suppose Φ has p distinct roots $g_1, \dots, g_p$, all with $|g_i| > 1$. Then $1/\Phi(x) = \sum_{k \geqslant 0} \psi_k x^k$ for $|x| < \min_i |g_i|$, with

$$\psi_k = \sum_{i=1}^{p} c_i g_i^{-k}, \qquad c_i = \prod_{l \neq i} \left(1 - g_i/g_l\right)^{-1}, \tag{18}$$

and $|\psi_k| \leqslant \big(\sum_i |c_i|\big)\varrho^k$ with $\varrho = \max_i |g_i|^{-1} < 1$. Consequently $y_t = \mu + \sum_{k\geqslant 0} \psi_k \varepsilon_{t-k}$, with $\mu = \beta_0/\Phi(1)$, converges in L^2 and is the unique weakly stationary solution of (14).

Proof. Since the roots are distinct, $\Phi(x) = \prod_i (1 - x/g_i)$ and the partial fraction expansion $1/\Phi(x) = \sum_i c_i/(1 - x/g_i)$ holds with c_i obtained by multiplying through by $(1 - x/g_i)$ and setting $x = g_i$, which gives $c_i = \prod_{l\neq i}(1 - g_i/g_l)^{-1}$. Expanding each factor as a geometric series, valid for $|x| < |g_i|$, and collecting powers gives (18) and the bound.

Absolute summability of $\{\psi_k\}$ makes the series square summable, so it converges in L^2. Applying $\Phi(\mathrm{B})$ term by term, the coefficient of ε_{t-k} in $\Phi(\mathrm{B})\sum_j \psi_j \varepsilon_{t-j}$ is the kth coefficient of $\Phi(x)\sum_j \psi_j x^j = 1$, hence 1 for $k = 0$ and 0 otherwise; the interchange is legitimate because $\Phi(\mathrm{B})$ is a finite linear combination and L^2 limits are preserved by such. So $\Phi(\mathrm{B})(y_t - \mu) = \varepsilon_t$, and $\Phi(\mathrm{B})\mu = \mu\Phi(1) = \beta_0$.

For uniqueness, let d_t be the difference of two weakly stationary solutions, so $\Phi(\mathrm{B})d_t = 0$ and $\mathsf{E}[d_t^2] \leqslant C$ free of t. Multiplying the truncated series $\sum_{k\leqslant K} \psi_k x^k$ by $\Phi(x)$ leaves 1 plus terms in $x^{K+1}, \dots, x^{K+p}$ whose coefficients are sums of at most p products $\psi_k\beta_j$ with $K - p < k \leqslant K$ (the coefficient of x^m is $-\sum_{m-p\leqslant k\leqslant K} \psi_k\beta_{m-k}$), hence bounded by $p\max_j|\beta_j|\sum_i |c_i|\varrho^{K-p+1} = C'\varrho^K$

with $C' = p \max_j |\beta_j| \sum_i |c_i| \varrho^{1-p}$. Applying the same identity with B in place of x to d_t,

$$0 = \sum_{k \leqslant K} \psi_k \, \Phi(\mathrm{B}) d_{t-k} = d_t + \sum_{j=K+1}^{K+p} e_{K,j} \, d_{t-j}, \qquad |e_{K,j}| \leqslant C' \varrho^K,$$

so $\mathsf{E}[d_t^2]^{1/2} \leqslant p C' \varrho^K C^{1/2} \to 0$ as $K \to \infty$ by the triangle inequality in L^2, forcing $d_t = 0$ almost surely. □

Remark. For $p = 1$ the condition is $|g_1| = |\beta_1|^{-1} > 1$, that is $|\beta_1| < 1$. The requirement is on the roots, not on the coefficients: an AR(2) with $\beta_1 = 1.02$ and $\beta_2 = -0.24$ is stationary, because $\Phi(x) = 1 - 1.02x + 0.24x^2$ has both roots outside the unit circle, while its first coefficient exceeds one.

Definition 10.21 (ψ-coefficients)**.** The representation $y_t = \mu + \sum_{k \geqslant 0} \psi_k \varepsilon_{t-k}$ with $\psi_0 = 1$ is the **ψ-coefficient representation**. For AR(1), $\psi_k = \beta_1^k$; for MA(q), $\psi_k = -\theta_k$ for $1 \leqslant k \leqslant q$ and 0 beyond.

4.1 Partial autocorrelation

Definition 10.22 (Partial autocorrelation)**.** For a weakly stationary series for which the covariance matrix of $(y_{t-1}, \dots, y_{t-k})$ is nonsingular, let $(\alpha_{1,k}, \dots, \alpha_{k,k})$ be the (unique) coefficients of the projection of $y_t - \mu$ on $\{y_{t-1} - \mu, \dots, y_{t-k} - \mu\}$, that is the minimiser of $\mathsf{E}[\{y_t - \mu - \sum_{j \leqslant k} a_j (y_{t-j} - \mu)\}^2]$. The **partial autocorrelation** at lag k is $\alpha_{k,k}$.

Theorem 10.23 (The partial correlogram of an AR(p) cuts off)**.** Let $\{y_t\}$ be the stationary AR(p) of Theorem 10.20. Then $(\alpha_{1,k}, \dots, \alpha_{k,k}) = (\beta_1, \dots, \beta_p, 0, \dots, 0)$ for every $k \geqslant p$. In particular $\alpha_{p,p} = \beta_p$ and $\alpha_{k,k} = 0$ for $k > p$.

Proof. Write $x_t = y_t - \mu$, so $x_t = \sum_{j \leqslant p} \beta_j x_{t-j} + \varepsilon_t$. Fix $k \geqslant p$ and let $a = (\beta_1, \dots, \beta_p, 0, \dots, 0)$. The residual of that candidate is exactly ε_t, and $x_{t-j} \in \mathcal{L}_{t-1}$ for $j \geqslant 1$ while ε_t is independent of $\mathcal{L}_{t-1}$ with mean zero, so $\mathsf{E}[\varepsilon_t x_{t-j}] = 0$ for $j = 1, \dots, k$. Those are the normal equations of the projection in L^2 (with $X = (x_{t-1}, \dots, x_{t-k})'$, $\boldsymbol{\Gamma} = \mathsf{E}[XX']$ and $\boldsymbol{\gamma} = \mathsf{E}[X x_t]$, $\mathsf{E}[(x_t - a'X)^2] = a'\boldsymbol{\Gamma}a - 2a'\boldsymbol{\gamma} + \gamma(0)$, whose gradient vanishes exactly when $\boldsymbol{\Gamma}a = \boldsymbol{\gamma}$), whose solution is unique when the covariance matrix of $(x_{t-1}, \dots, x_{t-k})$ is nonsingular. It is: a singular covariance matrix would

make some combination $\sum_{j\leqslant k} b_j x_{t-j}$ with $b \neq \mathbf{0}$ vanish almost surely. Since $\operatorname{cov}(x_{t-j}, \varepsilon_{t-l}) = \sigma^2\psi_{l-j}$, which is 0 for $l < j$ and σ^2 for $l = j$, taking the covariance of that combination with ε_{t-1} gives $b_1\sigma^2 = 0$, then with ε_{t-2} gives $b_2\sigma^2 = 0$, and so on: the system is triangular and forces every $b_j = 0$. □

Remark. The correlogram and the partial correlogram are complementary identifiers: an MA(q) has $\rho_k = 0$ beyond lag q and a partial correlogram that decays without cutting off, an AR(p) has $\alpha_{k,k} = 0$ beyond lag p and a correlogram that decays without cutting off. Neither cuts off for an ARMA with both orders positive and Φ, Θ without a common root. Only the two cut-off statements are proved here (Proposition 10.19 and Theorem 10.23); the statements that the other identifier does not cut off are not.

Proposition 10.24 (Yule–Walker equations)**.** The coefficients of Definition 10.22 solve the linear system

$$\rho_l = \sum_{j=1}^{k} \alpha_{j,k}\, \rho_{|l-j|}, \qquad l = 1, \dots, k. \tag{19}$$

Proof. The normal equations of the projection in Definition 10.22 (the vanishing of the gradient in the proof of Theorem 10.23) are $\mathsf{E}[\{x_t - \sum_j a_j x_{t-j}\}\, x_{t-l}] = 0$ for $l = 1, \dots, k$, where $x_t = y_t - \mu$; that is $\gamma(l) = \sum_j a_j \gamma(l-j)$, and dividing by $\gamma(0)$, which is positive as a diagonal entry of the nonsingular covariance matrix of Definition 10.22, gives (19). □

Remark. Two sample versions of $\hat{\alpha}_{k,k}$ are in use and they are not the same number. Substituting the sample autocorrelations r_k into (19) and solving, which is what `pacf` does, forms every inner product from all T observations; regressing y_t on an intercept and $y_{t-1}, \dots, y_{t-k}$ forms them from the $T - k$ dates at which every regressor exists. The difference is an end effect, of order T^{-1} (not proved here).

Theorem 10.25 (General ψ-weight forecast; FREES §8.5.2)**.** For any process with the representation $y_t = \mu + \sum_{k \geqslant 0} \psi_k \varepsilon_{t-k}$ and $\sum_k \psi_k^2 < \infty$,

$$\mathsf{E}[y_{T+l} \mid \mathcal{L}_T] = \mu + \sum_{k \geqslant l} \psi_k \varepsilon_{T+l-k}, \qquad \operatorname{var}\big(y_{T+l} - \mathsf{E}[y_{T+l} \mid \mathcal{L}_T]\big) = \sigma^2 \sum_{k=0}^{l-1} \psi_k^2. \tag{20}$$

Proof. Split the sum at $k = l$. The terms with $k < l$ carry innovations dated after T, which are independent of $\mathcal{L}_T$ with mean zero; the terms with $k \geqslant l$ carry innovations dated at or before T, which lie in $\mathcal{L}_T$. The forecast error is therefore $\sum_{k<l} \psi_k \varepsilon_{T+l-k}$, a sum of uncorrelated terms. □

Remark (FREES §8.5.2)**.** FREES writes the forecast-error variance as $\sigma^2 \sum_{k=1}^{l-1} \psi_k^2$; the sum starts at $k = 0$, as in its own interval and in Theorem 10.25.

Example 10.4 (Hong Kong exchange rates; FREES §8.6, Tables 8.6–8.8)**.** The daily rates of Unit 9, differenced ($T = 501$), with ARIMA$(2, 1, 0)$ and ARIMA$(0, 1, 2)$ fits:

```
hk <- read.csv("data/frees/csv/HKExchange.csv")$EXHKUS; dh <- diff(hk)
round(c(sd_diff = sd(dh), se_rk = 1 / sqrt(length(dh))), 4)
```

```
## sd_diff   se_rk
##  0.0020  0.0447
```

```
round(acf(dh, lag.max = 10, plot = FALSE)$acf[2:11], 3)
```

```
##  [1]  0.078 -0.151 -0.038 -0.001  0.095 -0.005  0.051 -0.012
##  [9]  0.084 -0.001
```

```r
round(pacf(hk, lag.max = 3, plot = FALSE)$acf[, 1, 1], 3)
```

```
## [1]  0.988 -0.034  0.051
```

```r
fa <- arima(dh, order = c(2, 0, 0))
fm <- arima(dh, order = c(0, 0, 2))
tr <- function(f) round(c(coef(f), coef(f) / sqrt(diag(f$var.coef)), s =
 ↪  sqrt(f$sigma2)), 5)
tab <- rbind(ar2 = tr(fa), ma2 = tr(fm))
colnames(tab) <- c("coef1", "coef2", "intercept", "t1", "t2", "t_int", "s"); tab
```

```
##       coef1   coef2 intercept    t1     t2  t_int       s
## ar2 0.08991 -0.1574   0.00003 2.039 -3.572 0.3186 0.00193
## ma2 0.09181 -0.1611   0.00003 2.067 -3.469 0.3231 0.00193
```

```r
lb <- function(f) sapply(c(2, 4, 6, 8, 10), function(K) Box.test(residuals(f), K,
 ↪  "Ljung-Box")$statistic)
round(rbind(ar2 = lb(fa), ma2 = lb(fm)), 4)
```

```
##     X-squared X-squared X-squared X-squared X-squared
## ar2    0.0065    0.5674     6.350     10.45     16.33
## ma2    0.0110    0.2872     6.656     11.34     17.69
```

FREES's AR(2) is reproduced up to rounding: coefficients 0.0899 and -0.157 with $t = 2.04$ and -3.57 (exact likelihood here), against FREES's 0.0900 and -0.158 with $t = 2.03$ and -3.57. The MA(2) gives $t = 2.07$ and -3.47 (R writes $+\theta$ where FREES writes $-\theta$), against FREES's 2.08 and 3.66 in absolute value, so its second t is not reproduced. Both fits give $s = 0.00193$, and the Ljung–Box statistics agree with FREES's Table 8.8 to within 0.05. The partial autocorrelation 0.988 at lag 1 of the undifferenced series is FREES's sign of nonstationarity. FREES's Table 8.6 gives -0.151 as the lag-2 autocorrelation of the differences, which the series reproduces; its text drops the sign. The -0.158 is the lag-2 partial autocorrelation (FREES's Table 8.7) and FREES's AR(2) coefficient. FREES's §8.6 quotes the quadratic trend's $R^2 = 86.2\%$, not the 92.9% of its §7.2 (Unit 9).

Example 10.5 (An autoregression for NYSE trading volume; ISLR §10.5.2, §10.9.6)**.** ISLR's `NYSE`: daily standardised log trading volume v_t, Dow Jones return r_t and log volatility z_t, December 1962 to December 1986, trained before 1980 and tested after. Least squares of v_t on an intercept and the five lags of all three series, the lagged design of ISLR (10.21), then with the day of the week, and the "straw man" $\hat{v}_t = v_{t-1}$; test $R^2 = 1 - \mathrm{MSE}/V_0$ with V_0 the test variance:

```
xd <- scale(data.matrix(NYSE[, c("DJ_return", "log_volume", "log_volatility")]))
lagm <- function(x, k) rbind(matrix(NA, k, ncol(x)), x[1:(nrow(x) - k), ])
af <- data.frame(log_volume = xd[, "log_volume"], L1 = lagm(xd, 1),
  L2 = lagm(xd, 2), L3 = lagm(xd, 3), L4 = lagm(xd, 4), L5 = lagm(xd, 5))[-(1:5), ]
itr <- NYSE$train[-(1:5)]; yte <- af[!itr, "log_volume"]; V0 <- var(yte)
r2 <- function(p) 1 - mean((p - yte)^2) / V0
afd <- data.frame(day = NYSE[-(1:5), "day_of_week"], af)
fit5 <- lm(log_volume ~ ., data = af[itr, ]); fitd <- lm(log_volume ~ ., data =
 ↪  afd[itr, ])
round(c(train = sum(itr), test = sum(!itr), ar5 = r2(predict(fit5, af[!itr, ])),
        ar5_day = r2(predict(fitd, afd[!itr, ])), straw_man = r2(af[!itr,
         ↪  "L1.log_volume"])), 3)
```

```
##     train      test       ar5   ar5_day straw_man
##  4276.000  1770.000     0.413     0.460     0.181
```

ISLR's test R^2 of 0.41 for the autoregression, 0.46 with the day of the week, and 0.18 for the straw man are reproduced. The straw man is Unit 9's random-walk forecast without drift; the autoregression is a conditional least squares fit of Definition 10.8 with fifteen lagged regressors, assessed out of sample as in Unit 9.

5 Moving Averages and Exponential Smoothing

FREES §9.1–9.2. SRM learning outcomes 3(b), 3(c).

Definition 10.26 (Running average; FREES (9.1))**.** The running average of length k is $\hat{s}_t = k^{-1}\sum_{j=0}^{k-1} y_{t-j}$, and the forecast it supplies is $\hat{y}_{T+l} = \hat{s}_T$ for every $l \geqslant 1$.

Proposition 10.27 (Update and weighted least squares; FREES (9.2)–(9.3))**.**

(i) $\hat{s}_t = \hat{s}_{t-1} + (y_t - y_{t-k})/k$.

(ii) $\hat{s}_T$ is the weighted least squares estimate of β_0 in $y_t = \beta_0 + \varepsilon_t$ with weights $v_t = \mathbb{1}\{t > T-k\}$.

(iii) Under the model of (ii), $\operatorname{var}(\hat{s}_t) = \sigma^2/k$.

Proof. (i) $k\hat{s}_t - k\hat{s}_{t-1} = y_t - y_{t-k}$, the two windows differing in one term at each end. (ii) The weighted criterion is $\sum_{t>T-k}(y_t - b_0)^2$, a quadratic in b_0 with derivative $-2\sum_{t>T-k}(y_t - b_0)$, hence minimised at the mean of the last k observations. (iii) The average of k uncorrelated terms. □

Proposition 10.28 (A running average lags a linear trend; FREES §9.1)**.** Let $k \geqslant 2$. If $y_t = \beta_0 + \beta_1 t + \varepsilon_t$ then

$$\mathsf{E}[\hat{s}_t] = \beta_0 + \beta_1\Big(t - \frac{k-1}{2}\Big), \qquad \mathsf{E}\big[\hat{s}_t^{(2)}\big] = \beta_0 + \beta_1\big(t - (k-1)\big), \tag{21}$$

$\hat{s}^{(2)}$ being the running average of $\hat{s}^{(1)} = \hat{s}$. Consequently $2\hat{s}_T^{(1)} - \hat{s}_T^{(2)}$ is unbiased for the trend value at T and $b_{1,T} = 2(\hat{s}_T^{(1)} - \hat{s}_T^{(2)})/(k-1)$ is unbiased for β_1.

Proof. $\mathsf{E}[\hat{s}_t] = k^{-1}\sum_{j<k}\{\beta_0 + \beta_1(t-j)\} = \beta_0 + \beta_1 t - \beta_1 k^{-1}\sum_{j<k} j$, and $\sum_{j<k} j = k(k-1)/2$. Applying the same operator again shifts the argument by another $(k-1)/2$. The two combinations follow by solving the pair of linear equations: $2\,\mathsf{E}[\hat{s}^{(1)}] - \mathsf{E}[\hat{s}^{(2)}] = \beta_0 + \beta_1 t$ and $\mathsf{E}[\hat{s}^{(1)}] - \mathsf{E}[\hat{s}^{(2)}] = \beta_1(k-1)/2$. □

Remark. Under a linear trend with positive slope, a running average is in expectation too low by $(k-1)/2$ times the slope. Reading the smoothed curve as an estimate of the current level is therefore wrong whenever there is a trend, and double smoothing exists to correct exactly that.

5.1 Exponential smoothing

Definition 10.29 (Exponential smoothing; FREES (9.4)–(9.6))**.** For $w \in (0,1)$ and a starting value $\hat{s}_0$,

$$\hat{s}_t = (1-w)y_t + w\,\hat{s}_{t-1}, \tag{22}$$

and the forecast is $\hat{y}_{T+l} = \hat{s}_T$ for every $l \geqslant 1$. The smoothing parameter is chosen to minimise the sum of squared one-step errors $\mathrm{SS}(w) = \sum_{t=2}^{T}(y_t - \hat{s}_{t-1})^2$. From here on $\hat{s}_t$ (with $\hat{s}_t^{(2)}$, $b_{0,t}$, $b_{1,t}$ built from it) denotes the exponentially smoothed series, not the running average of Definition 10.26.

Proposition 10.30 (Unrolled form and discounted least squares; FREES §9.2)**.** (i)

$$\hat{s}_T = (1-w)\sum_{j=0}^{T-1} w^j y_{T-j} + w^T \hat{s}_0.$$

Thus the weights on the observations sum to $1-w^T$ and decay geometrically.

(ii) The weighted least squares estimate of β_0 in $y_t = \beta_0 + \varepsilon_t$ with weights $v_t = w^{T-t}$ is $\sum_{t\leqslant T} w^{T-t}y_t / \sum_{t\leqslant T} w^{T-t}$, which is $\hat{s}_T$ up to the starting value and the normalisation $1-w^T$.

(iii) Under white noise the estimate in (ii) has variance tending to $\sigma^2(1-w)/(1+w)$ as $T \to \infty$, so smaller w tracks the data and larger w smooths it.

Proof. (i) Substitute (22) into itself T times. (ii) Differentiating $\sum_t w^{T-t}(y_t - b_0)^2$ in b_0 gives the stated ratio; the denominator is $(1-w^T)/(1-w)$, so the ratio is $(1-w)\sum_j w^j y_{T-j}/(1-w^T)$. (iii) The variance is $\sigma^2 \sum_t w^{2(T-t)}/(\sum_t w^{T-t})^2 \to \sigma^2(1-w)^2/(1-w^2)$, which is $\sigma^2(1-w)/(1+w)$. □

Theorem 10.31 (The model under which exponential smoothing is optimal (beyond the syllabus))**.** Let $\{y_t\}$ satisfy $\Delta y_t = \varepsilon_t - \theta\varepsilon_{t-1}$, the ARIMA$(0,1,1)$ model, with $0 < \theta < 1$ and a starting value $y_0 \in \mathcal{L}_0$ (for instance a constant), so that $y_t \in \mathcal{L}_t$ for $t \geqslant 0$. Then the minimum mean squared error one-step forecast satisfies the exponential smoothing recursion with $w = \theta$:

$$\hat{s}_T := \mathsf{E}[y_{T+1} \mid \mathcal{L}_T] = (1-\theta)y_T + \theta\,\hat{s}_{T-1}. \tag{23}$$

Proof. Taking conditional expectations in $y_{T+1} = y_T + \varepsilon_{T+1} - \theta\varepsilon_T$ and using $\mathsf{E}[\varepsilon_{T+1} \mid \mathcal{L}_T] = 0$ gives $\hat{s}_T = y_T - \theta\varepsilon_T$. Applied one step earlier the same identity reads $\hat{s}_{T-1} = y_{T-1} - \theta\varepsilon_{T-1}$, so

the one-step error is

$$y_T - \hat{s}_{T-1} = y_T - y_{T-1} + \theta\varepsilon_{T-1} = \varepsilon_T - \theta\varepsilon_{T-1} + \theta\varepsilon_{T-1} = \varepsilon_T.$$

Substituting $\varepsilon_T = y_T - \hat{s}_{T-1}$ into $\hat{s}_T = y_T - \theta\varepsilon_T$ gives $\hat{s}_T = y_T - \theta(y_T - \hat{s}_{T-1})$, which is (23). □

Remark. Exponential smoothing is therefore not a heuristic: it is the optimal forecast of a nonstationary process whose differences are MA(1), and the smoothing parameter is that model's moving average coefficient. The same calculation explains the flat forecast function: $\mathsf{E}[y_{T+l} \mid \mathcal{L}_T] = \hat{s}_T$ for every $l \geqslant 1$, since the extra innovations all have conditional mean zero.

Proposition 10.32 (Double exponential smoothing; FREES §9.2)**.** Let $\hat{s}_t^{(1)} = (1-w)y_t + w\hat{s}_{t-1}^{(1)}$ and $\hat{s}_t^{(2)} = (1-w)\hat{s}_t^{(1)} + w\hat{s}_{t-1}^{(2)}$, started in the infinite past, and let $y_t = \beta_0 + \beta_1 t + \varepsilon_t$ with $\mathsf{E}\,\varepsilon_t = 0$. Then

$$\mathsf{E}\,\hat{s}_t^{(1)} = \beta_0 + \beta_1 t - \frac{w}{1-w}\beta_1, \qquad \mathsf{E}\,\hat{s}_t^{(2)} = \beta_0 + \beta_1 t - \frac{2w}{1-w}\beta_1,$$

so $b_{0,t} = 2\hat{s}_t^{(1)} - \hat{s}_t^{(2)}$ and $b_{1,t} = \{(1-w)/w\}(\hat{s}_t^{(1)} - \hat{s}_t^{(2)})$ are unbiased for the level $\beta_0 + \beta_1 t$ and

the slope, and the forecast is $\hat{y}_{t+l} = b_{0,t} + b_{1,t}\, l$.

Proof. With the infinite past, $\hat{s}_t^{(1)} = (1-w)\sum_{j\geqslant 0} w^j y_{t-j}$, so $\mathsf{E}\,\hat{s}_t^{(1)} = (1-w)\sum_j w^j\{\beta_0+\beta_1(t-j)\} = \beta_0 + \beta_1 t - \beta_1(1-w)\sum_j jw^j$, and $\sum_{j\geqslant 0} jw^j = w/(1-w)^2$ (differentiate the geometric series). The same operator applied to the linear function $\mathsf{E}\,\hat{s}_t^{(1)}$ shifts it by another $w\beta_1/(1-w)$. Then $2\,\mathsf{E}\,\hat{s}_t^{(1)} - \mathsf{E}\,\hat{s}_t^{(2)} = \beta_0 + \beta_1 t$ and $\mathsf{E}\,\hat{s}_t^{(1)} - \mathsf{E}\,\hat{s}_t^{(2)} = w\beta_1/(1-w)$. □

Proposition 10.33 (Holt's method contains double exponential smoothing; FREES §9.3)**.** Holt's recursions $b_{0,t} = (1-w_1)y_t + w_1(b_{0,t-1}+b_{1,t-1})$ and $b_{1,t} = (1-w_2)(b_{0,t}-b_{0,t-1}) + w_2 b_{1,t-1}$, started at the double-smoothing values, reproduce the $b_{0,t}, b_{1,t}$ of Proposition 10.32 at every t when $w_1 = w^2$ and $w_2 = 2w/(1+w)$.

Proof. Write $\alpha = 1-w$, $s_t = \hat{s}_t^{(1)}$, $u_t = \hat{s}_t^{(2)}$, and use $u_t = \alpha s_t + w u_{t-1}$. Then $b_{0,t} = 2s_t - u_t = (1+w)s_t - wu_{t-1}$ and $b_{1,t} = (\alpha/w)(s_t - u_t) = \alpha(s_t - u_{t-1})$. **Level.** From $b_{0,t-1} = 2s_{t-1} - u_{t-1}$ and $b_{1,t-1} = (\alpha/w)(s_{t-1} - u_{t-1})$, $b_{0,t-1} + b_{1,t-1} = (2+\alpha/w)s_{t-1} - (1+\alpha/w)u_{t-1} = \{(1+w)s_{t-1} - u_{t-1}\}/w$, since $w + \alpha = 1$. So $(1-w^2)y_t + w^2(b_{0,t-1} + b_{1,t-1}) = (1+w)\alpha y_t +$

$w(1+w)s_{t-1} - wu_{t-1} = (1+w)s_t - wu_{t-1} = b_{0,t}$. **Slope.** With $1 - w_2 = \alpha/(1+w)$, $b_{0,t} - b_{0,t-1} = (1+w)s_t - wu_{t-1} - 2s_{t-1} + u_{t-1} = (1+w)s_t - 2s_{t-1} + \alpha u_{t-1}$, and

$$\frac{\alpha}{1+w}\{(1+w)s_t - 2s_{t-1} + \alpha u_{t-1}\} + \frac{2w}{1+w} \cdot \frac{\alpha}{w}(s_{t-1} - u_{t-1}) = \alpha s_t + \frac{\alpha(\alpha-2)}{1+w}u_{t-1} = \alpha(s_t - u_{t-1}),$$

as $\alpha - 2 = -(1+w)$; this is $b_{1,t}$. Induction on t from the common start finishes. □

Remark (FREES §9.3)**.** FREES states the equivalence with $w_1 = w_2 = 2w/(1+w)$; the level parameter is w^2, and with FREES's choice the two procedures differ.

Example 10.6 (Medical component of the CPI; FREES §9.1–9.2, Figures 9.2–9.3)**.** FREES's quarterly index, 1947 to 2007 Q1, as quarterly percentage changes ($T = 240$); exponential smoothing started at $\hat{s}_0 = y_1$, and SS(w) over a grid:

```
mc <- read.csv("data/frees/csv/MedCPISmooth.csv")
pm <- mc$PerMEDCPI[-1]
round(c(T = length(pm), first = mc$value[1], last = tail(mc$value, 1),
        quarterly_growth = (tail(mc$value, 1) / mc$value[1])^(1 / 240) - 1), 4)
```

```
##                T            first             last
##         240.0000          13.3000         345.9940
## quarterly_growth
##           0.0137
```

```
es <- function(y, w, s0 = y[1]) { s <- numeric(length(y)); prev <- s0
  for (t in seq_along(y)) { s[t] <- (1 - w) * y[t] + w * prev; prev <- s[t] }; s }
SSw <- function(w) { s <- es(pm, w); sum((pm[-1] - head(s, -1))^2) }
c(file_minus_manual = max(abs(es(pm, 0.2) - mc$MCPISMw_2[-1])),
  w_min = optimize(SSw, c(0.01, 0.99))$minimum)
```

```
## file_minus_manual             w_min
##         0.0008891         0.5487669
```

```
set.seed(3); yh <- cumsum(rnorm(60, 0.5)); w <- 0.7
s1 <- es(yh, w); s2 <- es(s1, w); B0 <- 2 * s1 - s2; B1 <- (1 - w) / w * (s1 - s2)
holt <- function(y, w1, w2) { b0 <- B0[1]; b1 <- B1[1]; out <- matrix(NA, length(y),
 ↪  2)
  for (t in 2:length(y)) { nb0 <- (1 - w1) * y[t] + w1 * (b0 + b1)
    b1 <- (1 - w2) * (nb0 - b0) + w2 * b1; b0 <- nb0; out[t, ] <- c(b0, b1) }; out }
```

```
hc <- holt(yh, w^2, 2 * w / (1 + w)); hf <- holt(yh, 2 * w / (1 + w), 2 * w / (1 +
 ↪  w))
signif(c(holt_w2_minus_double = max(abs(hc[-1, ] - cbind(B0, B1)[-1, ])),
         frees_choice_minus_double = max(abs(hf[-1, ] - cbind(B0, B1)[-1, ]))), 3)
```

```
##      holt_w2_minus_double frees_choice_minus_double
##                  7.11e-15                  2.22e+00
```

The index rose from 13.3 to 346.0, a quarterly growth of 1.37% (FREES 1.36%). The file's smoothed series agree with the recursion to its printed digits, and SS(w) is smallest near $w = 0.55$, FREES's "$w \approx 0.50$". Holt's method with $w_1 = w^2$ reproduces double smoothing to rounding error; with FREES's $w_1 = 2w/(1+w)$ it does not. Figure 10.2 shows the smoothed series and SS(w).

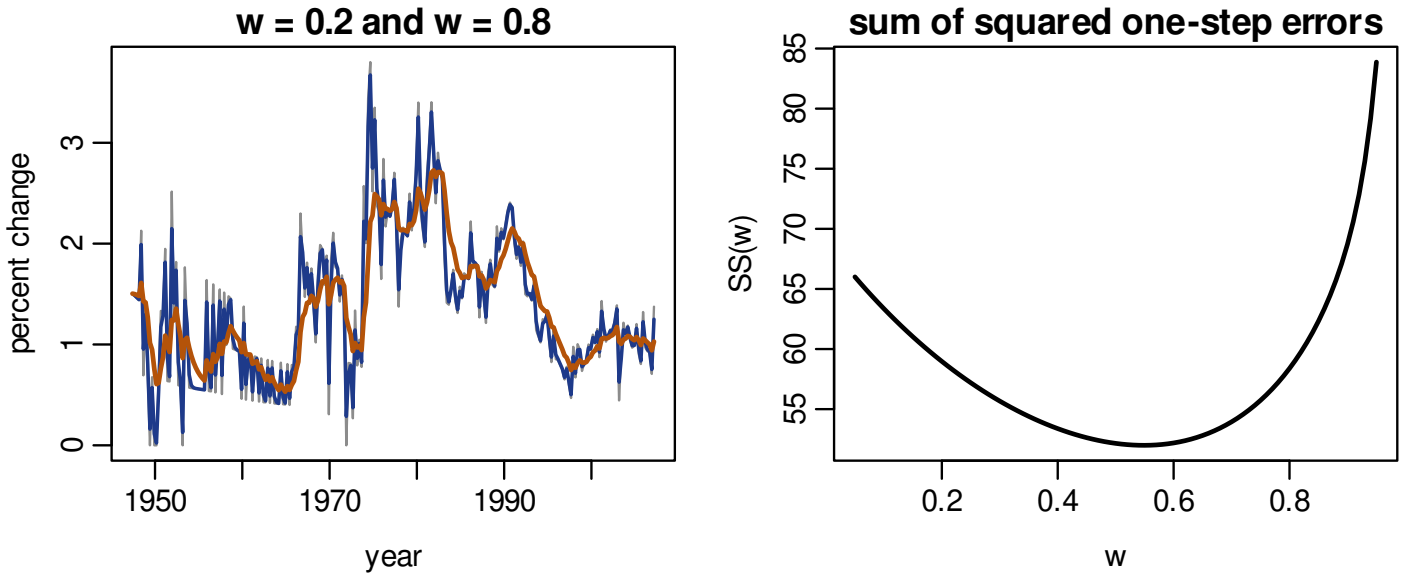


Figure 10.2: Quarterly percentage changes of the medical CPI with exponential smoothing at $w = 0.2$ and 0.8 (FREES Figure 9.2), and SS(w) (FREES Figure 9.3).

6 Seasonal Time Series Models

FREES §9.3. SRM learning outcome 3(b).

Proposition 10.34 (Trigonometric seasonals; FREES (9.7))**.** $a\sin(ft+b) = c_1\sin(ft) + c_2\cos(ft)$ with $c_1 = a\cos b$, $c_2 = a\sin b$, so a sum of m harmonics $f_i = 2\pi i/s$ (seasonal base s) enters a regression as $2m$ explanatory variables. For s even, $m \leqslant s/2$, and at $m = s/2$ the sine column $\sin(\pi t)$ vanishes at integer t, so only $s-1$ seasonal columns are available, as many as $s-1$ indicators.

Proof. The angle-sum identity $\sin(x+y) = \sin x\cos y + \cos x\sin y$. At $f = 2\pi(s/2)/s = \pi$, $\sin(\pi t) = 0$ for integer t. Harmonics with $i > s/2$ add nothing, because at integer t $\sin\{2\pi(s-i)t/s\} = -\sin(2\pi it/s)$ and $\cos\{2\pi(s-i)t/s\} = \cos(2\pi it/s)$. □

Example 10.7 (Cost of prescription drugs; FREES §9.3 (fit) and Table 9.1 (autocorrelations))**.** Monthly percentage changes of FREES's cost per claim (Unit 9), $T = 67$:

```
rx <- read.csv("data/frees/csv/PrescriptionDrug.csv"); yr <- rx$RATEC_C[-1]; tt <-
 ↪  rx$TIME[-1]
m1 <- lm(yr ~ sin(2 * pi * tt / 12) + cos(2 * pi * tt / 12))
m2 <- update(m1, . ~ . + sin(4 * pi * tt / 12) + cos(4 * pi * tt / 12))
round(coef(summary(m1))[, 1:3], 4)
```

```
##                      Estimate Std. Error t value
## (Intercept)            1.2217     0.2325   5.254
## sin(2 * pi * tt/12)   -1.6596     0.3269  -5.077
## cos(2 * pi * tt/12)    0.6538     0.3299   1.982
```

```
round(c(s_e = summary(m1)$sigma, R2_m1 = summary(m1)$r.squared, R2_m2 =
 ↪  summary(m2)$r.squared,
        partial_F = anova(m1, m2)$F[2],
        F_from_rounded_R2 = ((0.336 - 0.315) / 2) / ((1 - 0.336) / 62)), 4)
```

```
##              s_e            R2_m1            R2_m2
##           1.8973           0.3148           0.3365
##        partial_F F_from_rounded_R2
##           1.0143           0.9804
```

```
round(acf(yr, lag.max = 18, plot = FALSE)$acf[-1], 2)
```

```
##  [1]  0.08  0.10 -0.12 -0.11 -0.32 -0.33 -0.23  0.07 -0.08
## [10]  0.25  0.24  0.31 -0.01  0.14 -0.09 -0.08 -0.25 -0.18
```

FREES's fit is reproduced except for the sine coefficient, -1.6596, which FREES prints as -1.6956; its standard error 0.3269 and $t = -5.08$ are those of -1.6596. The partial F for the second harmonic is 1.014, p well above 0.05 (FREES's 0.98 comes from the rounded R^2's), so $m = 1$. The autocorrelations are FREES's Table 9.1 except at lags 7, 9 and 15 (-0.23, -0.08, -0.09, against FREES's -0.29, 0.08, -0.10); those at lags 5–6, 10–12 and 17 exceed $2/\sqrt{67} = 0.24$ in absolute value or nearly so.

Definition 10.35 (Seasonal autoregression; FREES (9.8))**.** With seasonal base s, the SAR(P) model is

$$y_t = \beta_0 + \beta_1 y_{t-s} + \beta_2 y_{t-2s} + \cdots + \beta_P y_{t-Ps} + \varepsilon_t, \tag{24}$$

which omits the intervening lags that an AR(Ps) would include.

Proposition 10.36 (SAR(1) is s interleaved AR(1) chains; FREES §9.3)**.** Let $\{y_t\}$ be the weakly stationary solution of the SAR(1) model $y_t = \beta_0 + \beta_1 y_{t-s} + \varepsilon_t$ with $|\beta_1| < 1$. Then each of the

s subseries $\{y_{ms+i}\}_m$, $i = 1, \ldots, s$, is a stationary AR(1) with coefficient β_1, the s subseries are independent of one another, and the autocorrelation function of $\{y_t\}$ is

$$\rho_k = \begin{cases} \beta_1^{k/s}, & s \mid k, \\ 0, & \text{otherwise.} \end{cases} \tag{25}$$

Proof. Reading (24) along a fixed residue class i modulo s gives exactly Definition 10.2 with the innovations $\{\varepsilon_{ms+i}\}_m$, which are independent and identically distributed; different residue classes use disjoint sets of innovations, hence are independent. Theorem 10.3 applied within a class gives $\rho_{ms} = \beta_1^m$, and two observations in different classes are independent, so their correlation is 0. □

Remark. The correlogram of a SAR(1) has spikes at $s, 2s, \ldots$ and nothing between, which is what distinguishes it from an AR(s): the latter fills in the intervening lags. A monthly series whose correlogram is large at 12 and 24 and negligible elsewhere is asking for the seasonal model, not the twelfth-order one, which would cost eleven extra parameters.

Example 10.8 (A seasonal autoregression; FREES §9.3)**.** FREES's SAR(2) with seasonal base $s = 6$, by conditional least squares:

```
nr <- length(yr); sar <- lm(yr[13:nr] ~ yr[7:(nr - 6)] + yr[1:(nr - 12)])
round(c(coef(summary(sar))[, 1], coef(summary(sar))[, 3], s_e = summary(sar)$sigma),
 ↪  4)
```

```
##     (Intercept)  yr[7:(nr - 6)] yr[1:(nr - 12)]
##          1.2191         -0.2867          0.3120
##     (Intercept)  yr[7:(nr - 6)] yr[1:(nr - 12)]
##          3.0001         -1.9093          2.0949
##             s_e
##          2.1563
```

FREES's $1.2191 - 0.2867\, y_{t-6} + 0.3120\, y_{t-12}$, t-statistics 3.00, -1.91, 2.09 and residual standard deviation $s_e = 2.156$ are reproduced; twelve observations are lost to the lags.

Definition 10.37 (Holt–Winters; FREES §9.3)**.** With smoothing parameters w_1, w_2, w_3 and

seasonal base s, the level, slope and seasonal estimates are updated by

$$b_{0,t} = (1 - w_1)(y_t - \hat{S}_{t-s}) + w_1(b_{0,t-1} + b_{1,t-1}), \tag{26}$$

$$b_{1,t} = (1 - w_2)(b_{0,t} - b_{0,t-1}) + w_2 b_{1,t-1}, \tag{27}$$

$$\hat{S}_t = (1 - w_3)(y_t - b_{0,t}) + w_3 \hat{S}_{t-s}, \tag{28}$$

and the forecast is $\hat{y}_{T+l} = b_{0,T} + b_{1,T} l + \hat{S}_{T+l-s\lceil l/s \rceil}$.

Proposition 10.38 (Holt–Winters reduces to what it generalises)**.** Dropping (28) and setting $\hat{S} \equiv 0$ leaves Holt's method for a local linear trend; setting in addition $b_{1,t} \equiv 0$ and dropping (27) leaves (22) with $w = w_1$.

Proof. Immediate substitution. □

7 Unit Root Tests

FREES §9.4. SRM learning outcome 3(a).

Proposition 10.39 (FREES's encompassing model; FREES §9.4)**.** In $y_t = \mu_0 + \phi(y_{t-1} - \mu_0) + \mu_1\{\phi + (1-\phi)t\} + \varepsilon_t$: $\phi = 1$ gives the random walk $y_t = \mu_1 + y_{t-1} + \varepsilon_t$; $\mu_1 = 0$, $|\phi| < 1$ gives the AR(1) with $\beta_0 = \mu_0(1-\phi)$; $\phi = 0$ gives the linear trend $\mu_0 + \mu_1 t + \varepsilon_t$. Subtracting y_{t-1} gives $\Delta y_t = \beta_0 + (\phi - 1)y_{t-1} + \beta_1 t + \varepsilon_t$ with $\beta_0 = \mu_0(1-\phi) + \phi\mu_1$ and $\beta_1 = \mu_1(1-\phi)$.

Proof. Substitute each value of ϕ (and μ_1) and collect terms: the right side is $\mu_0(1-\phi) + \phi\mu_1 + \phi y_{t-1} + \mu_1(1-\phi)t + \varepsilon_t$. □

Definition 10.40 (Dickey–Fuller regression; FREES (9.9)–(9.10))**.** Least squares is applied to

$$\Delta y_t = \beta_0 + (\phi - 1)\, y_{t-1} + \beta_1 t + \varepsilon_t, \tag{29}$$

and t_{DF} is the t ratio on y_{t-1}. The hypothesis $\phi = 1$ is a random walk and $\phi < 1$ a stationary alternative, so the test is one-sided to the left. The **augmented** version adds $\sum_{j \leqslant p} c_j \Delta y_{t-j}$ to

the right of (29) to absorb serial correlation in the errors.

Theorem 10.41 (The rate and the shape both change at the boundary)**.** Let $y_t = y_{t-1} + \varepsilon_t$ with $y_0 = 0$ and $\mathsf{E}[\varepsilon_t^4] < \infty$, and fit y_t on y_{t-1} without intercept, giving $\hat{\phi} = \sum_t y_{t-1} y_t / \sum_t y_{t-1}^2$. Then

$$\frac{1}{T}\sum_{t=1}^{T} y_{t-1}\varepsilon_t = \frac{1}{2}\Big(\frac{y_T^2}{T} - \frac{1}{T}\sum_{t=1}^{T}\varepsilon_t^2\Big) \xrightarrow{d} \frac{\sigma^2}{2}(\chi_1^2 - 1), \tag{30}$$

and

$$\mathsf{E}\Big[\frac{1}{T^2}\sum_{t=1}^{T} y_{t-1}^2\Big] = \frac{\sigma^2(T-1)}{2T} \longrightarrow \frac{\sigma^2}{2}. \tag{31}$$

So $T(\hat{\phi}-1)$, not $\sqrt{T}(\hat{\phi}-1)$, is the quantity of order one, and its numerator has a limit law that is not normal: it is skewed, with median at $\sigma^2(\chi^2_{1,0.5} - 1)/2 = -0.273\sigma^2$ and no mass below $-\sigma^2/2$.

Proof. From $y_t = y_{t-1} + \varepsilon_t$, squaring gives $y_t^2 = y_{t-1}^2 + 2y_{t-1}\varepsilon_t + \varepsilon_t^2$; summing over $t \leqslant T$ and telescoping gives $y_T^2 = 2\sum_t y_{t-1}\varepsilon_t + \sum_t \varepsilon_t^2$, which is the first equality in (30). Now $y_T/\sqrt{T} =$

$T^{-1/2}\sum_t \varepsilon_t \xrightarrow{d} N(0,\sigma^2)$ by the prerequisites' central limit theorem, so $y_T^2/T \xrightarrow{d} \sigma^2\chi_1^2$ by the continuous mapping theorem; and $T^{-1}\sum_t \varepsilon_t^2 \xrightarrow{p} \sigma^2$ by the prerequisites' lemma **variance of a stationary average** applied to the iid sequence $\{\varepsilon_t^2\}$, which the fourth moment makes square integrable. Slutsky's theorem combines the two.

For (31), $\mathsf{E}[y_{t-1}^2] = (t-1)\sigma^2$ and $\sum_{t=1}^{T}(t-1) = T(T-1)/2$.

For the order of $T(\hat\phi - 1) = N_T/D_T$, with $N_T = T^{-1}\sum_t y_{t-1}\varepsilon_t$ and $D_T = T^{-2}\sum_t y_{t-1}^2$: by Markov's inequality and (31), $\mathsf{P}\{D_T > M\} \leqslant \sigma^2/(2M)$. For a lower bound, the Cauchy–Schwarz inequality gives $D_T \geqslant S_T^2$ with $S_T = T^{-3/2}\sum_{t=1}^{T} y_{t-1} = T^{-3/2}\sum_{j=1}^{T-1}(T-j)\varepsilon_j$, whose variance $\sigma^2(T-1)(2T-1)/(6T^2)$ tends to $\sigma^2/3$. Each weight $(T-j)T^{-3/2}$ is at most $T^{-1/2}$ and $\operatorname{var}(S_T) \geqslant \sigma^2/8$ for $T \geqslant 2$, so the Lindeberg sum for $S_T/\operatorname{var}(S_T)^{1/2}$ at level η is at most $8\sigma^{-2}\,\mathsf{E}[\varepsilon_1^2;\, |\varepsilon_1| > c\eta\sqrt{T}] \leqslant 8\sigma^{-2}\,\mathsf{E}[\varepsilon_1^4]/(c^2\eta^2 T) \to 0$, with $c = \sigma/\sqrt{8}$, and the Lindeberg–Feller theorem of the prerequisites gives $S_T \xrightarrow{d} N(0,\sigma^2/3)$. Hence $\mathsf{P}\{D_T < \delta\} \leqslant \mathsf{P}\{|S_T| < \sqrt{\delta}\} \to \mathsf{P}\{|N(0,\sigma^2/3)| < \sqrt{\delta}\} \leqslant \{6\delta/(\pi\sigma^2)\}^{1/2}$, which is small for small δ. Since $\mathsf{P}\{|N_T/D_T| > M\} \leqslant \mathsf{P}\{|N_T| > M\delta\} + \mathsf{P}\{D_T < \delta\}$ and $\mathsf{P}\{|N_T/D_T| < \epsilon\} \leqslant \mathsf{P}\{|N_T| < \epsilon M\} + \mathsf{P}\{D_T > M\}$, and the limit law in (30) has a continuous distribution function, $T(\hat\phi - 1)$ is bounded in probability (for every $\eta > 0$ some

M gives $\mathsf{P}\{|T(\hat{\phi}-1)| > M\} < \eta$ for all large T) without tending to 0 in probability; in particular $\sqrt{T}(\hat{\phi}-1) = T^{-1/2} \cdot T(\hat{\phi}-1) \xrightarrow{p} 0$. □

Remark. The exact limit law of t_{DF} is **not** derived here. Obtaining it requires the joint limit of the numerator and the denominator, and the denominator's limit is a functional of a Brownian path, which the prerequisites do not construct. What Theorem 10.41 establishes, in the simplest case (no intercept, no trend), is the two facts that matter for using the test: the estimator converges at rate T rather than $\sqrt{T}$, and the regression standard error, itself of order T^{-1}, adapts to this, so t_{DF} stays of order one (this last fact is not part of Theorem 10.41 and is not proved here); and the limit of the numerator is not normal, so normal critical values have no justification. With an intercept or a trend, as in Definition 10.40, the analogous facts hold but are not proved here. The critical values are obtained by simulation instead.

Example 10.9 (Labor force participation rates; FREES §9.4, Table 9.2)**.** The Dickey–Fuller and augmented statistics for FREES's $T = 31$ rates, and the 10% points of t_{DF} under a random walk of the same length, by simulation:

```r
lf <- read.csv("data/frees/csv/LaborForcePR.csv")$MSC6U; dl <- diff(lf); Tl <- length(lf)
tdf <- function(x, p, trend) { d <- diff(x); n <- length(x); idx <- (p + 2):n
  X <- cbind(ylag = x[idx - 1]); if (trend) X <- cbind(X, t = idx)
  if (p > 0) for (j in 1:p) X <- cbind(X, d[idx - 1 - j])
  coef(summary(lm(d[idx - 1] ~ X)))[2, 3] }
round(rbind(no_trend = sapply(0:2, tdf, x = lf, trend = FALSE),
            trend = sapply(0:2, tdf, x = lf, trend = TRUE)), 3)
```

```
##             [,1]   [,2]   [,3]
## no_trend -1.614 -1.816 -1.736
## trend    -0.266 -0.037  0.421
```

```r
set.seed(94)
crit <- t(replicate(20000, { z <- cumsum(rnorm(Tl)); c(tdf(z, 0, FALSE), tdf(z, 0, TRUE)) }))
round(apply(crit, 2, quantile, 0.10), 3)
```

```
## [1] -2.621 -3.206
```

FREES's six statistics are reproduced; none is below the 10% points, so a random walk is not rejected. The simulated points, -2.62 and -3.21 at $T = 31$, are within 0.03 of FREES's -2.624 and

-3.228; the normal 10% point -1.28 would be badly wrong (the simulation above; Theorem 10.41 shows, for the regression without intercept or trend, why normal theory does not apply).

8 ARCH/GARCH Models

FREES §9.5. SRM learning outcome 3(b).

Remark (FREES §9.5)**.** Many financial series show **volatility clustering**: periods of large changes followed by periods of small ones, as in the daily S&P 500 returns of 2000–2006 (FREES Figure 9.10). A constant variance cannot describe this, yet weak stationarity requires one. The models of this section resolve the conflict by letting the variance **conditional on the past**, $\sigma_t^2 = \operatorname{var}(\varepsilon_t \mid \varepsilon_{t-1}, \varepsilon_{t-2}, \ldots)$, change with time while the unconditional variance stays constant. Forecasting the variance, not only the level, matters in finance, for example in option pricing.

Definition 10.42 (ARCH and GARCH; FREES §9.5)**.** Let $\{z_t\}$ be independent standard normal. The GARCH$(1,1)$ model is

$$\varepsilon_t = \sigma_t z_t, \qquad \sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \delta \sigma_{t-1}^2, \tag{32}$$

with $\omega > 0$, $\alpha, \delta \geqslant 0$ and $\alpha + \delta < 1$, and with σ_t a function of $z_{t-1}, z_{t-2}, \ldots$ (a non-anticipative solution). The case $\delta = 0$ is ARCH(1). FREES writes w, γ_1, δ_1 for ω, α, δ, and $\gamma_1 + \delta_1 < 1$ for $\alpha + \delta < 1$.

Theorem 10.43 (A GARCH series is white noise in the weak sense (uncorrelated) with a non-constant conditional variance; FREES §9.5)**.** Suppose $\{\varepsilon_t\}$ solves (32) and is weakly stationary with $\mathsf{E}[\varepsilon_t^2] < \infty$. Then

$$\mathsf{E}[\varepsilon_t] = 0, \qquad \operatorname{cov}(\varepsilon_t, \varepsilon_{t+k}) = 0 \ (k \neq 0), \qquad \operatorname{var}(\varepsilon_t) = \frac{\omega}{1 - \alpha - \delta}, \tag{33}$$

while the conditional variance given the past is σ_t^2, which for $\alpha > 0$ is a non-degenerate random variable. Moreover $\{\varepsilon_t^2\}$ satisfies

$$\varepsilon_t^2 = \omega + (\alpha + \delta)\varepsilon_{t-1}^2 + \nu_t - \delta\nu_{t-1}, \qquad \nu_t = \varepsilon_t^2 - \sigma_t^2, \tag{34}$$

which has the form of an ARMA(1, 1); if moreover $\mathsf{E}[\varepsilon_t^4] < \infty$, its innovations ν_t have mean zero and are uncorrelated; for $\alpha = 0$ they are iid, and for $\alpha > 0$ they are not independent (not proved here).

Proof. σ_t is a function of $z_{t-1}, z_{t-2}, \ldots$ (Definition 10.42), hence is independent of z_t, so $\mathsf{E}[\varepsilon_t] = \mathsf{E}[\sigma_t]\,\mathsf{E}[z_t] = 0$ and, for $k > 0$, $\mathsf{E}[\varepsilon_t\varepsilon_{t+k}] = \mathsf{E}[\varepsilon_t\sigma_{t+k}]\,\mathsf{E}[z_{t+k}] = 0$, $\varepsilon_t\sigma_{t+k}$ being determined by innovations dated before $t+k$. Taking expectations in (32), $\mathsf{E}[\varepsilon_t^2] = \mathsf{E}[\sigma_t^2]$ because $\mathsf{E}[z_t^2] = 1$, so weak stationarity gives $\mathsf{E}[\varepsilon^2] = \omega + \alpha\,\mathsf{E}[\varepsilon^2] + \delta\,\mathsf{E}[\varepsilon^2]$ and hence (33). If $\alpha > 0$, then $\sigma_t^2 = \omega + \sigma_{t-1}^2(\alpha z_{t-1}^2 + \delta)$ with $\sigma_{t-1}^2 \geqslant \omega > 0$ independent of z_{t-1}, so $\operatorname{var}(\sigma_t^2 \mid \sigma_{t-1}^2) = 2\alpha^2\sigma_{t-1}^4 > 0$ and σ_t^2 is not degenerate; if $\alpha = 0$, then $d_t = \sigma_t^2 - \omega/(1-\delta)$ satisfies $d_t = \delta^m d_{t-m}$ for every $m \geqslant 1$, and $\mathsf{E}\,|d_{t-m}| \leqslant \mathsf{E}[\sigma^2] + \omega/(1-\delta)$ is bounded by weak stationarity, so $\mathsf{E}\,|d_t| \leqslant \delta^m\{\mathsf{E}[\sigma^2] + \omega/(1-\delta)\} \to 0$, i.e. $\sigma_t^2 = \omega/(1-\delta)$, and $\nu_t = \{\omega/(1-\delta)\}(z_t^2 - 1)$ is iid.
For (34), substitute $\sigma_t^2 = \varepsilon_t^2 - \nu_t$ and $\sigma_{t-1}^2 = \varepsilon_{t-1}^2 - \nu_{t-1}$ into (32):

$$\varepsilon_t^2 - \nu_t = \omega + \alpha\varepsilon_{t-1}^2 + \delta(\varepsilon_{t-1}^2 - \nu_{t-1}),$$

which rearranges to the stated form. That $\nu_t = \sigma_t^2(z_t^2 - 1)$ has mean zero and is uncorrelated across t follows as above, $\mathsf{E}[z_t^2 - 1] = 0$ and σ_t^2 being determined by the past, the fourth moment making each ν_t square integrable. □

Definition 10.44 (ARCH(p) and GARCH(p, q); FREES §9.5)**.** With $\{z_t\}$ independent standard normal and $\varepsilon_t = \sigma_t z_t$, the GARCH($p, q$) model sets

$$\sigma_t^2 = \omega + \sum_{j=1}^{p} \alpha_j \varepsilon_{t-j}^2 + \sum_{j=1}^{q} \delta_j \sigma_{t-j}^2, \tag{35}$$

with $\omega > 0$, $\alpha_j, \delta_j \geqslant 0$ and $\sum_j \alpha_j + \sum_j \delta_j < 1$, and σ_t a function of $z_{t-1}, z_{t-2}, \ldots$; $q = 0$ is ARCH(p), and $p = q = 1$ is Definition 10.42 with $\alpha_1 = \alpha$, $\delta_1 = \delta$. FREES writes w and γ_j for ω and α_j, and $\gamma(1) + \delta(1) < 1$ for the condition on the sums. The ε_{t-j}^2 terms let a large shock raise the next variances; the σ_{t-j}^2 terms make the effect persist, as moving-average terms complement autoregressive ones.

Proposition 10.45 (The unconditional variance; FREES §9.5)**.** If $\{\varepsilon_t\}$ solves (35) and is weakly stationary with $\mathsf{E}[\varepsilon_t^2] < \infty$, then $\mathsf{E}[\varepsilon_t] = 0$, the ε_t are uncorrelated, and

$$\operatorname{var}(\varepsilon_t) = \frac{\omega}{1 - \sum_j \alpha_j - \sum_j \delta_j}.$$

Proof. As in the proof of Theorem 10.43, σ_t is independent of z_t, which gives the zero mean, the zero correlations and $\mathsf{E}[\varepsilon_t^2] = \mathsf{E}[\sigma_t^2]$. Taking expectations in (35) and writing v for the common value of $\mathsf{E}[\varepsilon_t^2] = \mathsf{E}[\sigma_t^2]$, $v = \omega + v\sum_j \alpha_j + v\sum_j \delta_j$, so $v = \omega/(1 - \sum_j \alpha_j - \sum_j \delta_j)$, positive because the two sums add to less than 1. □

Theorem 10.46 (Conditional normality produces unconditional fat tails)**.** For a weakly stationary ARCH(1) with $3\alpha^2 < 1$ and $\mathsf{E}[\varepsilon_t^4]$ finite and free of t,

$$\mathsf{E}[\varepsilon_t^4] = \frac{3\omega^2(1+\alpha)}{(1-\alpha)(1-3\alpha^2)}, \qquad \frac{\mathsf{E}[\varepsilon_t^4]}{\{\mathsf{E}[\varepsilon_t^2]\}^2} = \frac{3(1-\alpha^2)}{1-3\alpha^2} > 3 \quad \text{for } \alpha > 0. \tag{36}$$

Proof. $\mathsf{E}[\varepsilon_t^4] = \mathsf{E}[\sigma_t^4 z_t^4] = 3\,\mathsf{E}[\sigma_t^4]$, the fourth moment of a standard normal being 3 and σ_t independent of z_t. With $\delta = 0$,

$$\mathsf{E}[\sigma_t^4] = \mathsf{E}[(\omega + \alpha\varepsilon_{t-1}^2)^2] = \omega^2 + 2\omega\alpha\,\mathsf{E}[\varepsilon^2] + \alpha^2\,\mathsf{E}[\varepsilon^4],$$

and $\mathsf{E}[\varepsilon^2] = \omega/(1-\alpha)$ by (33). Substituting and solving for $\mathsf{E}[\varepsilon^4]$,

$$\mathsf{E}[\varepsilon^4](1-3\alpha^2) = 3\omega^2\left(1+\frac{2\alpha}{1-\alpha}\right) = \frac{3\omega^2(1+\alpha)}{1-\alpha}.$$

Dividing by $\{\omega/(1-\alpha)\}^2$ gives the kurtosis $3(1+\alpha)(1-\alpha)/(1-3\alpha^2)$, and $1-\alpha^2 > 1-3\alpha^2 > 0$ for $\alpha > 0$. □

Remark. The innovations are conditionally normal at every date, and the series is nonetheless leptokurtic: at $\alpha = 0.5$ the kurtosis is 9. Mixing normals of different variances produces heavy tails, as a plot of daily returns shows.

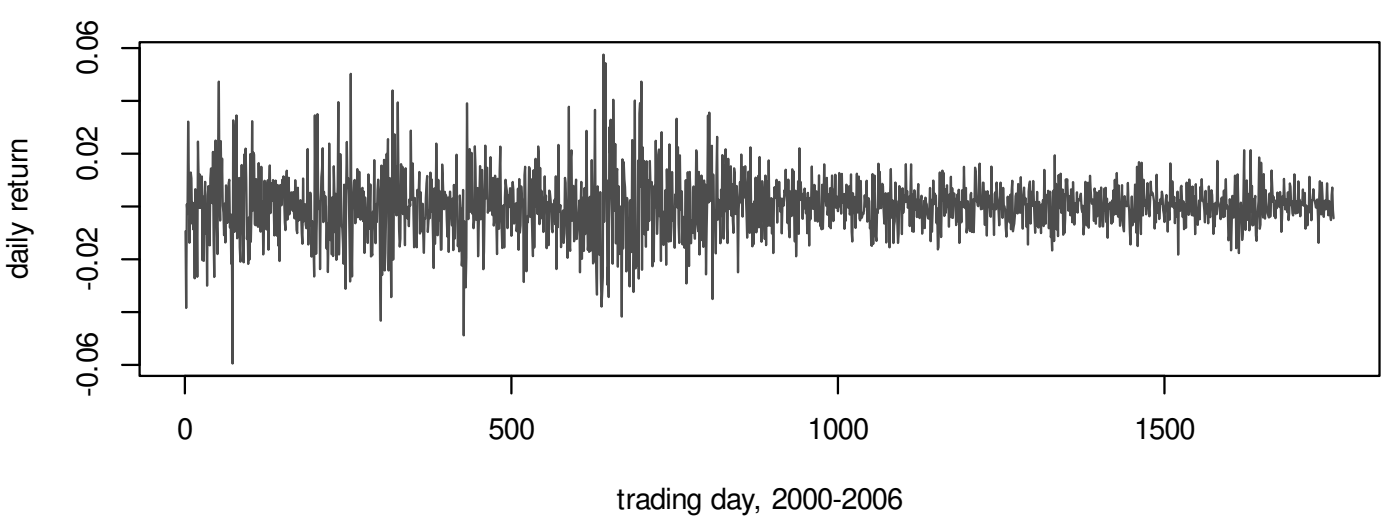
Definition 10.47 (Conditional likelihood)**.** Given $\varepsilon_1, \ldots, \varepsilon_n$ and a starting value σ_1^2, the recursion (32) determines $\sigma_2^2, \ldots, \sigma_n^2$, and the parameters are estimated by maximising

$$\ell(\omega, \alpha, \delta) = -\frac{1}{2}\sum_{t=2}^{n}\left\{\ln(2\pi\sigma_t^2) + \frac{\varepsilon_t^2}{\sigma_t^2}\right\}. \tag{37}$$

Remark. The likelihood (37) is not concave in (ω, α, δ) in general (not proved here), and its summands are dependent, so the maximum likelihood theorem of the prerequisites does not apply and the standard errors it would supply are not available here. The likelihood is maximised directly and checked against a package; the asymptotic theory of GARCH estimation is not developed here.

Example 10.10 (S&P 500 daily returns; FREES §9.5, Table 9.3)**.** FREES's model for the 1759 daily returns of 2000–2006: $y_t = \beta_0 + \varepsilon_t - \theta_1\varepsilon_{t-1} - \theta_2\varepsilon_{t-2}$ with GARCH$(1,1)$ errors, fitted by maximising the conditional Gaussian likelihood from three starting values:

```
spd <- read.csv("data/frees/csv/SP500Daily.csv")
ys <- spd$vwretd[spd$caldt >= 20000101 & spd$caldt <= 20061231]
par(mar = c(4.2, 4, 1.2, 0.5), cex = 0.8)
plot(ys, type = "l", col = "grey30", xlab = "trading day, 2000-2006",
     ylab = "daily return")
```




```
nll <- function(par) { b0 <- par[1]; th1 <- par[2]; th2 <- par[3]; om <- exp(par[4])
  al <- par[5]; de <- par[6]
  if (al < 0 || de < 0 || al + de >= 1) return(1e10)
  e1 <- 0; e2 <- 0; s2 <- var(ys); ll <- 0
  for (t in seq_along(ys)) { e <- ys[t] - b0 + th1 * e1 + th2 * e2
    s2 <- om + al * e1^2 + de * s2; ll <- ll + log(s2) + e^2 / s2; e2 <- e1; e1 <- e
    ↪ }
  0.5 * ll + 0.5 * length(ys) * log(2 * pi) }
```

```
starts <- list(c(5e-4, -0.04, -0.06, log(5e-7), 0.07, 0.92),
               c(3e-4, 0, 0, log(1e-6), 0.10, 0.85), c(4e-4, -0.02, -0.05,
                ↪  log(2e-6), 0.05, 0.90))
fits <- lapply(starts, function(st) {
  o <- optim(st, nll, control = list(maxit = 20000, reltol = 1e-12))
  optim(o$par, nll, method = "BFGS", control = list(maxit = 2000, reltol = 1e-14))})
best <- fits[[which.min(sapply(fits, `[[`, "value"))]]; pb <- best$par
print(sapply(fits, function(f) -f$value), digits = 8)
```

```
## [1] 5658.4674 5655.9386 5658.4674
```

```
signif(c(n = length(ys), beta0 = pb[1], theta1 = pb[2], theta2 = pb[3],
         omega = exp(pb[4]), alpha_on_e2 = pb[5], delta_on_sigma2 = pb[6],
         logLik = -best$value), 4)
```

```
##               n           beta0          theta1
##       1.759e+03       4.597e-04       3.897e-02
##          theta2           omega     alpha_on_e2
##       6.121e-02       5.537e-07       6.613e-02
## delta_on_sigma2          logLik
##       9.295e-01       5.658e+03
```

The first and third starts reach the same maximum and the second stops 2.5 lower, on the boundary $\alpha + \delta = 1$ where the penalty halts the search; several starts guard against such stops. The estimates match FREES's Table 9.3 in size: $b_0 = 0.00046$, $\hat{\omega} = 5.5 \times 10^{-7}$ (FREES 5.61×10^{-7}), the coefficient of ε_{t-1}^2 0.066 and that of σ_{t-1}^2 0.929, with log-likelihood 5658.47 against 5658.85 (FREES's start of the recursion is not stated). Two labels in FREES are inconsistent with its equation $\sigma_t^2 - \delta_1 \sigma_{t-1}^2 = w + \gamma_1 \varepsilon_{t-1}^2$: its table assigns 0.0667 to δ_1 and 0.9288 to γ_1, but the coefficient of σ_{t-1}^2 is the large one, and its θ_1, θ_2 carry the opposite sign to its moving average equation. $\hat{\alpha} + \hat{\delta} = 0.996$ lies near the boundary $\alpha + \delta = 1$, where the variance $\omega/(1 - \alpha - \delta)$ of Theorem 10.43 grows without bound. The plot, FREES's Figure 9.10, shows the volatility clustering the model is built for: the returns before 2003 swing far more widely than those after, with no trend in their level.

9 Regression Trees

ISLR §8.1.1, to tree pruning (which is Unit 11's), on both reading lists. SRM learning outcomes 4(a), 4(b).

Definition 10.48 (Partition predictor; ISLR §8.1.1)**.** Let $R_1, \dots, R_J$ be disjoint sets covering the predictor space. The associated **partition predictor** is

$$\hat{f}(\mathbf{x}) = \sum_{j=1}^{J} \hat{c}_j \, \mathbb{1}\{\mathbf{x} \in R_j\}, \tag{38}$$

constant on each region. A **regression tree** is a partition predictor whose regions are boxes obtained by successive splits on one coordinate at a time.

Remark (ISLR §8.1.1)**.** ISLR's vocabulary: the regions are the **terminal nodes** or **leaves** of the tree, the points along the tree where the predictor space is split are its **internal nodes**, and the segments joining the nodes are its **branches**; the tree is drawn upside down, the leaves at the bottom. Not every partition into boxes is a tree: ISLR's Figure 8.3 shows five boxes arranged like

a pinwheel, which no sequence of splits of the whole space produces, since the first split of a tree must cut the entire space in two along one coordinate.

Proposition 10.49 (The region constants are the region means)**.** For fixed regions each containing at least one observation, the values $\hat{c}_1, \dots, \hat{c}_J$ minimising $\sum_{i=1}^{n} \{y_i - \hat{f}(\mathbf{x}_i)\}^2$ are the region means $\hat{c}_j = n_j^{-1} \sum_{i:\mathbf{x}_i \in R_j} y_i$, where $n_j = |\{i : \mathbf{x}_i \in R_j\}|$, and the minimised value is $\sum_j \sum_{i \in R_j} (y_i - \hat{c}_j)^2$.

Proof. The criterion separates into J sums, the jth involving only c_j; each is minimised at the mean of its observations (Unit 2's least squares with an intercept only). □

Proposition 10.50 (A tree is a linear model on indicators)**.** Let $\mathbf{D}$ be the $n \times J$ matrix with $D_{ij} = \mathbb{1}\{\mathbf{x}_i \in R_j\}$. For fixed regions each containing at least one observation ($n_j \geqslant 1$ for every j), the fitted values of (38) are $\mathbf{D}(\mathbf{D}'\mathbf{D})^{-1}\mathbf{D}'\mathbf{Y}$, the orthogonal projection of $\mathbf{Y}$ on the column space of $\mathbf{D}$, and $\mathbf{D}'\mathbf{D} = \operatorname{diag}(n_1, \dots, n_J)$.

Proof. The regions are disjoint and cover the space, so each row of $\mathbf{D}$ has exactly one 1 and the columns are orthogonal with squared norms n_j. Least squares on $\mathbf{D}$ is Proposition 10.49, and Unit 2 identifies the fitted values with the projection. □

Remark. The columns of $\mathbf{D}$ are the seasonal indicators of Unit 9 with the calendar replaced by a partition of the predictor space; the identifiability discussion there applies verbatim, so (38) carries no separate intercept. What makes a tree different from an ordinary dummy regression is not the fit given the regions, which is a projection, but that the regions are chosen from the same data.

9.1 Recursive binary splitting

Remark. Minimising the residual sum of squares over all partitions into J boxes is not a computation that can be carried out. Even the restricted family in which each of p predictors is cut at one of $n-1$ places, applied recursively, has more members than can be enumerated: the first split alone admits $p(n-1)$ choices, and each subsequent split multiplies the count again. Trees are therefore grown greedily, one split at a time.

Definition 10.51 (Split; ISLR (8.2)–(8.3))**.** A split of a node R on coordinate j at cutpoint c produces $R_L = R \cap \{\mathbf{x} : x_j < c\}$ and $R_R = R \cap \{\mathbf{x} : x_j \geqslant c\}$. Recursive binary splitting chooses, at each node and among the (j, c) for which R_L and R_R both contain observations, the one minimising

$$\sum_{i \in R_L} (y_i - \bar{y}_L)^2 + \sum_{i \in R_R} (y_i - \bar{y}_R)^2, \tag{39}$$

and stops when a node has fewer than a preset number of observations.

Theorem 10.52 (What a split gains)**.** Let a node contain n observations with mean $\bar{y}$, and let a split send n_L of them to the left with mean $\bar{y}_L$ and $n_R = n - n_L$ to the right with mean $\bar{y}_R$, where $n_L, n_R \geqslant 1$. Then

$$\sum_{i} (y_i - \bar{y})^2 = \sum_{i \in R_L} (y_i - \bar{y}_L)^2 + \sum_{i \in R_R} (y_i - \bar{y}_R)^2 + \frac{n_L n_R}{n} (\bar{y}_L - \bar{y}_R)^2. \tag{40}$$

Minimising (39) is therefore the same as maximising $n_L n_R (\bar{y}_L - \bar{y}_R)^2 / n$, and the gain is zero exactly when the two region means coincide.

Proof. Split the left sum at $\bar{y}_L$: $\sum_{i\in R_L}(y_i-\bar{y})^2=\sum_{i\in R_L}(y_i-\bar{y}_L)^2+n_L(\bar{y}_L-\bar{y})^2$, the cross term vanishing because $\sum_{i\in R_L}(y_i-\bar{y}_L)=0$; similarly on the right. Adding,

$$\sum_{i}(y_i-\bar{y})^2=\sum_{i\in R_L}(y_i-\bar{y}_L)^2+\sum_{i\in R_R}(y_i-\bar{y}_R)^2+n_L(\bar{y}_L-\bar{y})^2+n_R(\bar{y}_R-\bar{y})^2.$$

Since $n\bar{y}=n_L\bar{y}_L+n_R\bar{y}_R$, one has $\bar{y}_L-\bar{y}=(n_R/n)(\bar{y}_L-\bar{y}_R)$ and $\bar{y}_R-\bar{y}=-(n_L/n)(\bar{y}_L-\bar{y}_R)$, so the last two terms are $\{n_Ln_R^2+n_Rn_L^2\}(\bar{y}_L-\bar{y}_R)^2/n^2=n_Ln_R(\bar{y}_L-\bar{y}_R)^2/n$. □

Corollary 10.53 (Cost of one node)**.** Sorting each predictor once, the best split of a node of size n over p predictors is found in $O(pn)$ further operations.

Proof. Sweeping the sorted values of one predictor from left to right, n_L increases by one at a time and $\sum_{i\in R_L}y_i$ is maintained by a single addition, so $\bar{y}_L$, $\bar{y}_R$ and the gain of Theorem 10.52 are each computed in a fixed number of operations per candidate cutpoint. There are $n-1$ cutpoints per predictor. □

Remark. The identity (40) is the one-way analysis of variance decomposition of Unit 3 with two groups. It says a split is judged by the separation of the two group means weighted by $n_L n_R/n$, so a small group must lie far from the rest to be split off.

Proposition 10.54 (Monotone invariance)**.** Replacing a predictor x_j by $g(x_j)$ for any strictly increasing g leaves every tree grown by Definition 10.51 unchanged, up to relabelling the cutpoints.

Proof. The candidate splits of coordinate j are determined by the ordering of the observed values, which g preserves, and the criterion (39) depends on the split only through which observations fall on each side. □

Remark. A tree therefore needs no decision about logarithms, standardisation or outlying predictor values, all of which change a linear fit. It pays for that with the two weaknesses of a piecewise-constant predictor: the fitted surface is a step function, so a genuinely linear relation is approximated by a staircase that needs many splits, and the prediction at any $\mathbf{x}$ outside the range of the training data equals the prediction at the nearest point of the box (each coordinate clamped to its observed range, since a cutpoint whose two children are both nonempty satisfies

$\min_j < c \leqslant \max_j$, the extremes of the observed x_j), so a tree cannot extrapolate at all.

Example 10.11 (Baseball salaries; ISLR §8.1.1, Figures 8.1–8.2)**.** ISLR's `Hitters` without missing salaries ($n = 263$), log(`Salary`) on `Years` and `Hits`; the tree grown by `tree` and its best three-leaf subtree, which is ISLR's:

```
suppressMessages(library(tree)); Hi <- na.omit(Hitters)
tH <- prune.tree(tree(log(Salary) ~ Years + Hits, data = Hi), best = 3); print(tH,
 ↪  digits = 5)
```

```
## node), split, n, deviance, yval
##       * denotes terminal node
##
## 1) root 263 207.15 5.927
##   2) Years < 4.5 90  42.35 5.107 *
##   3) Years > 4.5 173  72.70 6.354
##     6) Hits < 117.5 90  28.09 5.998 *
##     7) Hits > 117.5 83  20.88 6.740 *
```

```
leaf <- ifelse(Hi$Years < 4.5, "R1", ifelse(Hi$Hits < 117.5, "R2", "R3"))
round(rbind(mean_log_salary = tapply(log(Hi$Salary), leaf, mean),
```

```
    salary_dollars = 1000 * exp(tapply(log(Hi$Salary), leaf, mean))), 3)
```

```
##                          R1         R2        R3
## mean_log_salary       5.107      5.998      6.74
## salary_dollars  165139.341 402775.706 845296.05
```

```
ss <- function(v) sum((v - mean(v))^2); lsal <- log(Hi$Salary); L <- Hi$Years < 4.5
round(c(gain_first_split = ss(lsal) - ss(lsal[L]) - ss(lsal[!L]),
        formula = sum(L) * sum(!L) / length(L) * (mean(lsal[L]) -
         ↪  mean(lsal[!L]))^2), 4)
```

```
## gain_first_split          formula
##             92.1             92.1
```

The splits **Years** < 4.5 and **Hits** < 117.5 give leaf means 5.107, 5.998 and 6.740, and ISLR's prediction $e^{5.107}$ thousand dollars, \$165,174, for the first leaf (\$165,139 from the unrounded mean). The gain of the first split equals $n_L n_R(\bar{y}_L - \bar{y}_R)^2/n$, Theorem 10.52.

10 Computation in R

FREES chapters 8–9 scripts; ISLR §10.9.6 (beyond the syllabus).

Example 10.12 (Conditional least squares four ways; Definition 10.8)**.** On the inflation bond index: `lm`, the closed form of Proposition 10.9, `arima` with `method = "CSS"`, and the approximation r_1:

```
ym <- yb[-Tb]; yp <- yb[-1]
b1_closed <- sum((ym - mean(ym)) * (yp - mean(yp))) / sum((ym - mean(ym))^2)
acss <- arima(yb, order = c(1, 0, 0), method = "CSS")
print(c(lm = coef(mb)[[2]], closed_form = b1_closed, arima_css =
  coef(acss)[["ar1"]],
        r1 = rb[1], arima_default = coef(arima(yb, order = c(1, 0, 0)))[["ar1"]]),
        digits = 8)
```

```
##             lm   closed_form     arima_css            r1
##    0.81383964    0.81383964    0.81384350    0.81384766
## arima_default
##    0.85382567
```

All four agree to within 10^{-5} here; the end effect of Proposition 10.9 shows in b_0 (Example 10.2: least squares 0.4085 against $\bar{y}(1 - r_1) = 0.4179$), and the default `arima`, which maximises the exact likelihood, differs again.

Example 10.13 (Partial autocorrelations from the Yule–Walker system; Proposition 10.24)**.**

```
ra <- acf(yb, lag.max = 3, plot = FALSE)$acf[2:4]
yw <- sapply(1:3, function(k) { R <- toeplitz(c(1, ra)[1:k]); solve(R, ra[1:k])[k] })
reg <- sapply(1:3, function(k) { X <- sapply(1:k, function(j) yb[(k + 1 - j):(Tb - j)])
  tail(coef(lm(yb[(k + 1):Tb] ~ X)), 1) })
round(rbind(yule_walker = yw, pacf = pacf(yb, lag.max = 3, plot = FALSE)$acf[, 1, 1],
            lagged_regression = unname(reg)), 4)
```

```
##                    [,1]    [,2]   [,3]
## yule_walker      0.8138 -0.0886 0.2161
## pacf             0.8138 -0.0886 0.2161
## lagged_regression 0.8138 -0.1591 0.3198
```

`pacf` solves the Yule–Walker system in the sample autocorrelations; the lagged regression uses only the dates at which every lag exists, and differs by an end effect.

Example 10.14 (Portmanteau statistics; Theorem 10.13)**.** Ljung–Box by hand against `Box.test`, and the null distribution for white noise and for the residuals of a fitted AR(1), $T = 100$, $K = 10$, 5000 replications:

```
eh <- residuals(fa)[-(1:2)]; Kl <- 10; nT <- length(eh)
rk <- acf(eh, lag.max = Kl, plot = FALSE)$acf[-1]
c(by_hand = nT * (nT + 2) * sum(rk^2 / (nT - 1:Kl)),
  Box.test = unname(Box.test(eh, Kl, "Ljung-Box")$statistic))
```

```
##  by_hand Box.test
##    16.26    16.26
```

```
set.seed(105)
q <- replicate(5000, { z <- arima.sim(list(ar = 0.6), 100)
  c(white = Box.test(rnorm(100), 10, "Ljung-Box")$statistic,
    ar1_resid = Box.test(residuals(arima(z, order = c(1, 0, 0))), 10,
     ↪  "Ljung-Box")$statistic) })
round(rbind(mean = rowMeans(q), size_chisq10 = rowMeans(q > qchisq(0.95, 10)),
```

```
    size_chisq9 = rowMeans(q > qchisq(0.95, 9))), 3)
```

```
##                white.X-squared ar1_resid.X-squared
## mean                    10.115                9.013
## size_chisq10             0.063                0.037
## size_chisq9              0.087                0.055
```

For white noise the mean is near 10 and χ^2_{10} rejects 6.3% of the time at $T = 100$; for fitted residuals the mean is near 9 and only $\chi^2_{K-p} = \chi^2_9$ keeps the size near 5%.

11 Exercises

Exercise 10.1 (FREES exercise 8.3)**.** Using Unit 2's simple regression with y_{t-1} in place of x_t, $t = 2, \ldots, T$: (a) give b_1 exactly; (b) give b_0 exactly; (c) show $b_0 \approx \bar{y}(1 - r_1)$.

Solution. (a), (b) Proposition 10.9. (c) The same proposition: $b_1 \approx r_1$ and $\bar{y}_{\pm} \approx \bar{y}$ up to end effects, so $b_0 = \bar{y}_+ - b_1 \bar{y}_- \approx \bar{y}(1 - r_1)$. On the inflation bond index, $b_0 = 0.4085$ against $\bar{y}(1 - r_1) = 0.4179$ (Example 10.2). ■

Exercise 10.2 (FREES exercise 8.4)**.** For the AR(1) model show (a) $\sigma_y^2(1-\beta_1^2)=\sigma^2$; (b) $\operatorname{cov}(y_t,y_{t-1})=\beta_1\sigma_y^2$; (c) $\operatorname{cov}(y_t,y_{t-k})=\beta_1^k\sigma_y^2$; (d) $\rho_k=\beta_1^k$.

Solution. Theorem 10.3. Directly: (a) ε_t is uncorrelated with y_{t-1}, so taking variances in (1), $\sigma_y^2=\beta_1^2\sigma_y^2+\sigma^2$. (b), (c) Multiply $y_t-\mu=\beta_1(y_{t-1}-\mu)+\varepsilon_t$ by $y_{t-k}-\mu$ and take expectations: $\gamma(k)=\beta_1\gamma(k-1)$, since ε_t is uncorrelated with y_{t-k} for $k\geqslant 1$; iterate from $\gamma(0)=\sigma_y^2$. (d) Divide by σ_y^2. ■

Exercise 10.3 (FREES exercise 8.5)**.** For the AR(1) model, (a) use the chain rule to show $y_{T+k}-\hat{y}_{T+k}\approx\varepsilon_{T+k}+\beta_1\varepsilon_{T+k-1}+\cdots+\beta_1^{k-1}\varepsilon_{T+1}$; (b) show that its variance is about $\sigma^2\sum_{l=0}^{k-1}\beta_1^{2l}$.

Solution. Theorem 10.15: with the parameters known the chain rule gives $\hat{y}_{T+k}=\mu+\beta_1^k(y_T-\mu)$ and the error is exactly $\sum_{l<k}\beta_1^l\varepsilon_{T+k-l}$, of variance $\sigma^2\sum_{l<k}\beta_1^{2l}$; with estimated parameters the error differs by a term of order $T^{-1/2}$ (Theorem 10.10 and its proof), hence "$\approx$". ■

Exercise 10.4 (FREES exercise 8.6, part (c))**.** For the 503 daily returns of the S&P value-weighted index in 2005–2006 (Unit 9), compute the autocorrelations at lags 1 to 10 and say which

differ significantly from zero.

Solution.

```
r56 <- spd$vwretd[spd$caldt >= 20050101 & spd$caldt <= 20061231]
rr <- acf(r56, lag.max = 10, plot = FALSE)$acf[-1]
round(rbind(r_k = rr, z = rr * sqrt(length(r56))), 3)
```

```
##       [,1]   [,2]  [,3]   [,4]   [,5]   [,6]   [,7]   [,8]
## r_k -0.046 -0.096 0.019 -0.002 -0.004 -0.054 -0.035 -0.034
## z   -1.042 -2.152 0.431 -0.049 -0.088 -1.222 -0.788 -0.754
##       [,9] [,10]
## r_k -0.051 0.026
## z   -1.142 0.592
```

```
round(c(band = 2 / sqrt(length(r56)), Q_LB = unname(Box.test(r56, 10,
 ↪  "Ljung-Box")$statistic),
       p = Box.test(r56, 10, "Ljung-Box")$p.value), 4)
```

```
##    band    Q_LB       p
##  0.0892 10.3783  0.4080
```

By Corollary 10.7, under white noise each $|r_k|$ exceeds $2/\sqrt{503} = 0.089$ with probability about

0.045. Only $r_2 = -0.096$ ($z = -2.15$) crosses the band, about what ten looks at white noise would produce, and the Ljung–Box statistic for the ten lags jointly, 10.38, gives $p = 0.41$ against χ^2_{10} (Theorem 10.13): no autocorrelation is detected, as the market efficiency argument of FREES's exercise expects. ■

Exercise 10.5 (ISLR §8.4, exercise 1)**.** Draw a partition of a two-dimensional predictor space that recursive binary splitting could produce, with at least six regions, and the corresponding tree, labelling regions and cutpoints.

Solution. Split $X_1 < t_1$; on $\{X_1 < t_1\}$ split $X_2 < t_2$, giving R_1 below and, above, split $X_1 < t_3$ ($t_3 < t_1$) into R_2 and R_3; on $\{X_1 \geqslant t_1\}$ split $X_2 < t_4$, giving R_4 above and, below, split $X_1 < t_5$ ($t_5 > t_1$) into R_5 and then $X_2 < t_6$ on the rest into R_6, R_7. Every region is a box obtained by successive one-coordinate splits (Definition 10.51); Figure 10.3 draws both:

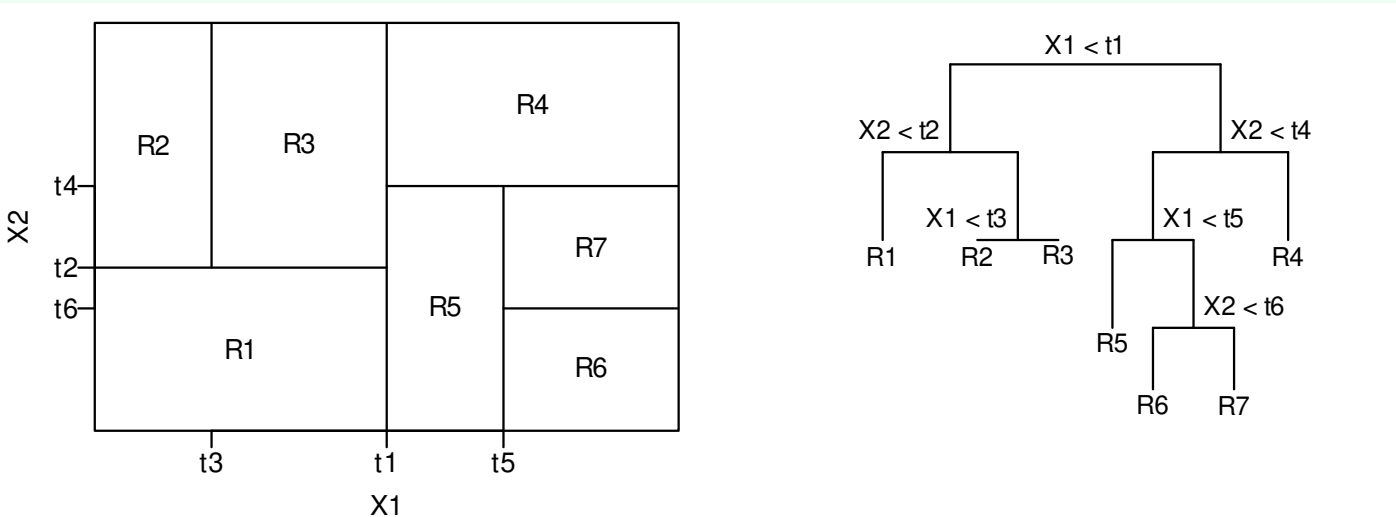


Figure 10.3: A seven-region partition from recursive binary splitting and its tree.

Exercise 10.6 (ISLR §8.4, exercise 4)**.** (a) Sketch the tree for the partition of ISLR Figure 8.14 (left): regions with means 15 on $\{X_1 < 1, X_2 \geqslant 1\}$, 3 on $\{X_1 < 0, X_2 < 1\}$, 0 on $\{0 \leqslant X_1 <$

$1, 0 \leqslant X_2 < 1\}$, 10 on $\{0 \leqslant X_1 < 1, X_2 < 0\}$ and 5 on $\{X_1 \geqslant 1\}$. (b) Draw the partition for the tree of Figure 8.14 (right): $X_2 < 1$, then $X_1 < 1$ with leaves -1.80 and 0.63; otherwise $X_2 < 2$, then $X_1 < 0$ with leaves -1.06 and 0.21, and leaf 2.49 for $X_2 \geqslant 2$.

Solution. (a) Root $X_1 < 1$: right leaf 5. Left: $X_2 < 1$; its right child is the leaf 15; its left child splits $X_1 < 0$ into the leaf 3 and a node $X_2 < 0$ with leaves 10 ($X_2 < 0$) and 0. (b) The line $X_2 = 1$ divides the plane; below it, $X_1 = 1$ separates -1.80 (left) from 0.63; above it, $X_2 = 2$ separates the strip $1 \leqslant X_2 < 2$, cut by $X_1 = 0$ into -1.06 and 0.21, from the half-plane $X_2 \geqslant 2$ with mean 2.49. Each leaf mean is the region mean (Proposition 10.49). ■

Exercise 10.7 (ISLR §10.10, exercise 9; beyond the syllabus)**.** Fit the lag-5 autoregression to `NYSE`, and refit it with a twelve-level month factor. Does the month improve the test R^2?

Solution.

```
mon <- factor(substr(as.character(NYSE$date), 6, 7))[-(1:5)]
afm <- data.frame(month = mon, af)
fitm <- lm(log_volume ~ ., data = afm[itr, ])
```

```
round(c(ar5 = r2(predict(fit5, af[!itr, ])), ar5_month = r2(predict(fitm, afm[!itr,
 ↪  ]))), 4)
```

```
##       ar5 ar5_month
##    0.4132    0.4170
```

The month factor changes the test R^2 in the third decimal only; the day of the week (Example 10.5) matters, the month does not. ■

Exercise 10.8 (ISLR §10.10, exercise 10; beyond the syllabus)**.** Fit the linear autoregression by "flattening" the 5×3 array of lagged values (ISLR §10.9.6 builds it as the input of a recurrent network), and compare its test R^2 with the `lm` fit.

Solution. The flattened array of the five lagged 3-vectors is the same set of fifteen regressors in another order, so least squares gives the same fit (Unit 2: fitted values depend on the column space only):

```
xr <- array(data.matrix(af[, -1]), c(nrow(af), 3, 5))[, , 5:1]
flat <- matrix(xr, nrow(af)); fitf <- lm(af$log_volume[itr] ~ flat[itr, ])
pf <- cbind(1, flat[!itr, ]) %*% coef(fitf)
```

```
round(c(flattened = r2(pf), lm = r2(predict(fit5, af[!itr, ]))), 6)
```

```
## flattened        lm
##    0.4132    0.4132
```

The two are identical. `lm` on a data frame keeps names and handles factors such as the day of the week; the flattened array is the input format ISLR's recurrent network needs. ■

Exercise 10.9 (SOA sample question 22)**.** For the stationary AR(1) $y_t = \beta_0 + \beta_1 y_{t-1} + \varepsilon_t$, which statement is NOT true? (A) β_0 must not equal 1. (B) $|\beta_1| < 1$. (C) If $\beta_1 = 0$ the model is white noise. (D) If $\beta_1 = 1$ it is a random walk. (E) Only y_{t-1} is used as a predictor for y_t.

Solution. Theorem 10.3 needs $|\beta_1| < 1$ (B) and puts no restriction on β_0, so (A) is false. (C), (D) are Proposition 10.4, and (E) is the definition. The answer is (A). ■

Exercise 10.10 (SOA sample question 58)**.** For $y_1, \ldots, y_6 = 31, 35, 37, 41, 45, 51$ from an AR(1), the approximation to conditional least squares is used. Calculate s^2: (A) 13; (B) 21; (C) 22; (D) 26; (E) 35.

Solution. The approximation of Proposition 10.9 is $b_1 = r_1$, $b_0 = \bar{y}(1 - r_1)$:

```
y6 <- c(31, 35, 37, 41, 45, 51); a6 <- y6 - mean(y6)
r1s <- sum(a6[-1] * a6[-6]) / sum(a6^2); b0s <- mean(y6) * (1 - r1s)
e6 <- y6[-1] - (b0s + r1s * y6[-6])
round(c(r1 = r1s, b0 = b0s, mean_e = mean(e6), s2 = sum((e6 - mean(e6))^2) / (6 -
  3),
        s2_least_squares = summary(lm(y6[-1] ~ y6[-6]))$sigma^2), 4)
```

```
##              r1              b0         mean_e
##          0.4466         22.1374          2.7824
##              s2 s2_least_squares
##         21.9707          1.5251
```

$r_1 = 117/262 = 0.4466$, $b_0 = 22.137$, the residuals average 2.78 (not 0, as the fit is not least squares) and (6) gives $s^2 = 21.97$: (C). Conditional least squares itself gives a much smaller s^2. ■

Exercise 10.11 (SOA sample question 64)**.** For a stationary AR(1) $y_t = \beta_0 + \beta_1 y_{t-1} + \varepsilon_t$, which is always true? (A) $\beta_0 \neq 0$; (B) $\beta_0 = 1$; (C) $\beta_1 = 0$; (D) $\beta_1 = 1$; (E) $|\beta_1| < 1$.

Solution. Theorem 10.3 and Proposition 10.4: stationarity requires $|\beta_1| < 1$, and β_0 is any constant: (E). ■

Exercise 10.12 (SOA sample question 46)**.** A series observed at $t = 0, \ldots, 100$ ends with $y_{97}, \ldots, y_{100} = 96.9, 98.1, 99.0, 100.2$, with $\hat{s}_{97}^{(1)}, \hat{s}_{98}^{(1)}, \hat{s}_{99}^{(1)} = 93.1, 94.1, 95.1$ and $\hat{s}_{97}^{(2)}, \hat{s}_{98}^{(2)} = 88.9, 89.9$ at $w = 0.8$. With F the exponential smoothing and G the double exponential smoothing forecast of y_{102} (forecasts to one decimal, the trend to three), calculate $|F - G|$: (A) 0.0; (B) 2.1; (C) 4.2; (D) 6.3; (E) 8.4.

Solution. By Definition 10.29, $\hat{s}_{100}^{(1)} = 0.2(100.2) + 0.8(95.1) = 96.1$, the forecast at every lead, so $F = 96.1$. Then $\hat{s}_{99}^{(2)} = 0.2(95.1) + 0.8(89.9) = 90.9$ and $\hat{s}_{100}^{(2)} = 0.2(96.1) + 0.8(90.9) = 91.9$. By Proposition 10.32, $b_{1,100} = (0.2/0.8)(96.1 - 91.9) = 1.05$ and $b_{0,100} = 2(96.1) - 91.9 = 100.3$, so $G = 100.3 + 2(1.05) = 102.4$ and $|F - G| = 6.3$: (D). ■